



Indira Gandhi National Open University
School of Journalism and New Media Studies

MDC-004

**DEVELOPMENT JOURNALISM
FOR SOCIAL CHANGE**

**DEVELOPMENT JOURNALISM FOR
SOCIAL CHANGE**



**School of Journalism and New Media Studies
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MDC-004 INTRODUCTION TO COURSE: DEVELOPMENT JOURNALISM FOR SOCIAL CHANGE

The Course titled Development Journalism for Social Change was designed and developed to introduce the development issues and their significance in bringing more comprehensive discussion through media. In the earlier, the government played a crucial role in the development. Later the government realized the significance of the participatory mode of development and the role of media. Media started regularly publishing on the issues of development. The media realized that the development is not only improving infrastructure; it is about addressing the needs of all and improving the standard of living among all. Media play a crucial role in bringing development issues for wider discussion and to facilitate political institutions to reflect the same in the policies. With this backdrop, the Course has divided into four blocks.

Block No. 1: Issues in Development Journalism-I

The first Block is “Issues in Development Journalism-I which covers development issues, governance issues, agriculture and rural issues and science and technology issues. India entered into the neoliberal economic model in 1991, and it decentralised the government structure, and the governance becomes prominent in the 21st Century. The role of media in governance is significant because governance encourages everyone in society to actively participate in improving all lives. The block has four Units.

Unit 1: Developmental Issues deals with this essential subject which has to be the primary thrust of journalism in developing societies. A journalist needs to learn the art of reporting and writing on issues related to the economy, social sector, environment, infrastructure etc., which are dealt with in this Unit.

Unit 2: Governance Issues provides a basic understanding of the issues related to governance. The development of any society depends on its legal and administrative frameworks of governance, which determine people’s well-being in a society. Understanding governance gives you an insight into shaping the content of a news story.

Unit 3: Agricultural and Rural Issues describes the importance of agriculture and rural development in a country with a largely agrarian economy. It is even argued that agriculture and rural reporting is the future of Indian journalism as our city-centric media makes inroads into rural areas.

Unit 4: Science & Technology Issues explains the concept and nature of science journalism, which is still in its infancy in India. With the industrial and technological development of the country, Science journalism is poised for new opportunities and challenges in future.

It will be helpful for you to develop a good understanding of these crucial areas and equip yourself with the requisite skills to pursue these beats with confidence.

Block No. 2: Health and Hygiene issues in Development Journalism-II

Block 2 is Health and Hygiene issues of Development Journalism –II. The entire Block covers broad topics like health and sanitation, education and media, media and environment, economy and finance and industrialisation and urbanisation.

Unit 5: Health and Sanitation, covers the sanitation situation of India compared with other countries. The Unit also describes the impact of open defecation among women and children, significantly impacting their health directly. Finally, the Unit insisted on the importance of media to cover health and sanitation issues.

Unit 6: Education and Media, discusses the significance and functions of education. It takes an overview of the education systems in pre-independent and contemporary India and examines some initiatives of using media and ICT for development at the school and higher education levels. The Unit started the significance of investing in education for human development in the country. The Unit also described the existing government programmes to improve education in the country. Finally, the Unit explained the role of media in education. It includes the role of technology and media in facilitating curriculum development; learning improves literacy, career guidance, increasing Gross Enrollment Ratio, and media role in Open and Distance learning (ODL). Thus Unit six covers various aspects of education as well as the role of media.

Unit 7: Media and Environment, examines various environmental issues such as global warming, pollution and wildlife conservation and some other emerging areas of concern. In this modern knowledge-society, media plays a vital role as a facilitator of development, disseminator of information, and change agent. The Unit suggests alternatives for environment protection to people and policy-makers. The Unit introduced the environmental issues and talked about environmental communication, environmental journalism, and functions of mass media towards the environment, media platforms, and environmental awareness. Media too actively involved in various environmental campaign and movements in India like the “save tiger campaign”. Finally, the Unit discussed international conventions and conferences on climate change.

Unit 8: Economy and Finance, exposes you to the requirements of a business/finance reporter. Economic journalism reports as well as guides the society on path of development. This Unit deals with different aspects of economic and financial reporting such as share market, mutual funds and how they have impacted financial markets. It also enlists the media skills required to be a good reporter/ writers on economic matters.

Unit 9: Industrialisation and Urbanisation, explain the concept of Industrialisation, Urbanisation and development, describe the environmental impacts of Industrialisation. The Unit also makes the learner to understand the concept of Special Economic Zones (SEZ). SEZ played a critical role in the economic development of the country and creating employment opportunities. At the same time, the studies clearly pointed out in improving the existing condition of SEZs. The Unit specifically discusses the socio-economic impact of industrialisation As a learner you must know what is Urbanisation as well as you need to identify the problems associated with Urbanisation. It was discussed in the Unit. The Unit further discussed the major Urban Development Projects in India and the role of Communication in Urban Development

Block No. 3: Media Planning for Development

Block 3 is about media Planning and Development. The following Units are part of the Block—Planning a Development Communication Campaign, Development Communication Campaign: Case Studies, Implementation of Development Communication Projects and Corporate Social Responsibility for Development.

Unit 10: Planning a Development Communication Campaign- The Unit first introduced the concepts of development communication. In this section, the course writer also introduced the goals of development communication. As we introduced at the beginning of the Course, there are various development theories; like modernisation theories, dependency theory, world-systems theory and the theories proposed by feminists. Along with these development theories, the primary paradigm of development communication emerged. This was discussed in this Unit elaborately. After discussing significant paradigms, the Unit introduced approaches to development communication campaigns. These approaches included the adoption ladder, integrated model, communication for development, right-based approach, micro-level approach and macro-level approach. Finally, the Unit talked about planning a development communication campaign. This includes social marketing, advocacy and entertainment education.

Unit 11: Development Communication Campaigns: Case Studies, introduced various case studies, and the Unit is titled Development Communication Campaign: Case Studies. After indicating the core concepts of development communication, including “A Campaign- Purpose, Goals and Meaningful Change, Broad types of campaign and steps to design campaign, the Unit discussed case studies related to development communication campaigns drawn from various parts of the world.

Unit 12: Implementation of Development Communication Project developed to make the learners understand the changing paradigms in development communication, identify the various steps involved in creating a development communication project, and understand the value of development communication and the implementation of development communication project.

Unit 13: Corporate Social Responsibility for Development, spells out the much needed and now, lawfully compelled activity of CSR. Corporate do not work in isolation. Other than employees and consumers, they also work with other businesses, government, local authorities, and the general public too. All these are the stakeholders, not shareholders. Corporate social responsibility (CSR) is a type of international private business self-regulation that aims to contribute to societal goals of a philanthropic, activist, or charitable nature by engaging in or supporting volunteering or ethically oriented practices. In this Unit we talk in detail about CSR and its connection to business. We'll also study the role of stakeholders and how a corporate fulfils its responsibilities towards them. Topics like corporate governance and its interrelation with CSR are also covered. You will see how Corporate Social Responsibility is a management concept whereby companies integrate social and environmental concerns in their businesses to keep up with their image of goodness.

Block No. 4: Participatory Development Communication

The last and final Block this Course is Block 4. In this Block, we have covered the following Units namely Media Literacy, Right to Information, Civic Journalism, Mobile Journalism, Community Media and Development.

Unit 14: Media Literacy, stresses the need for media education in the present globalised media environment. The unlimited avenues for information generation and dissemination have led to blinding glare and often less clarity about the true meaning and role of media in society. Media literacy helps in differentiating

between ‘reel’ and the ‘real’ and as such make better use of communication and media resources.

Unit 15: Right to Information, describes the historical evolution of the Right to Information in India and explains various provisions of the Act in detail. It analyses the impact of RTI and examines the issues related to the implementation of RTI in India.

Unit 16: Civic Journalism is a field that has expanded over the years to include citizen or public journalism as well. Earlier, it was only limited to covering issues related to the life of a common man. This unit details the features, strengths and limitations of civic reporting. It also covers citizen journalism and ethical aspects of citizen reporting.

Unit 17: Mobile Journalism contains detailed information on the techniques of using a mobile phone for journalism. For becoming a mobile journalist (MoJo), a Smartphone is an essential tool. In fact, using a mobile phone, you can cover news, features, interviews, etc., for any news organisation – be it a print publication, a television channel, radio station or web network.

Unit 18: Community Media and Development, forms of community media, the role of community media, characteristics of community media and few community media projects in India was discussed.

As you all are aware, we in IGNOU, write each programme under several courses. Each course has further been divided into several Blocks. Each Block has several Units. In each Unit, you will find some questions and activities after one or two headings and at the end of the Unit. Please try to attempt these questions and activities to check your level of understanding of the subject.

Happy learning

Ramesh Yadav

Block

1

ISSUES IN DEVELOPMENT JOURNALISM -I

UNIT 1

Developmental Issues

UNIT 2

Governance Issues

UNIT 3

Agricultural and Rural Issues

UNIT 4

Science & Technology Issues

UNIT 1 DEVELOPMENTAL ISSUES

Structure

- 1.0 Introduction
- 1.1 Learning Outcomes
- 1.2 Conceptualising Development
 - 1.2.1 Defining Development
 - 1.2.2 Development Communication
 - 1.2.3 Development Journalism
- 1.3 Major Developmental Issues/Areas
 - 1.3.1 The Economy
 - 1.3.2 Social Sector
 - 1.3.3 Government and Governance
 - 1.3.4 Environment
 - 1.3.5 Infrastructure
 - 1.3.6 Agriculture/Rural
- 1.4 Reporting Development
 - 1.4.1 Principles of good reporting
 - 1.4.2 Techniques of good reporting
- 1.5 Reporting Development for Different media
 - 1.5.1 Print
 - 1.5.2 Radio
 - 1.5.3 Television
 - 1.5.4 Films
 - 1.5.5 New Media
- 1.6 Let Us Sum Up
- 1.7 References and Further Readings
- 1.8 Answers to Check your Progress Exercises

1.0 INTRODUCTION

While development as phenomena is as old as human civilisation, development studies emerged after the Second World War. The concept of development has been associated with Westernisation, modernisation, progress, social change and sustainability. There have been paradigm shifts, and changes in the way development was conceived and theorised. These changes in definition and theory have impacted reporting of developmental issues in media which have been dealt with in this Block.

Over time, the reporting of development issues from a narrow focus on the economy and related fields has broadened to include the social sector, environment, telecommunications, policy and planning. Earlier reporting used to be by wire services for radio or print media which has widened to include new media. At the same time, principles and techniques of reporting on development are mostly the same, some differences, especially concerning new and digital media. There are various challenges in reporting developmental issues, which we shall discuss in this Unit.

1.1 LEARNING OUTCOMES

After studying this Unit you should be able to :

- differentiate between the concepts of development, development communication and development journalism;
- discuss important developmental issues;
- describe coverage of development related issues;
- analyse what makes for good reportage of developmental issues;
- examine reporting on developmental issues for legacy media as compared to new media;
- identify challenges to reporting on developmental issues; and
- report on major developmental issues.

1.2 CONCEPTUALISING DEVELOPMENT

The field of development and its offshoots - development communications and development journalism emerged following the Second World War. The Western world had witnessed unparalleled death and destruction during World War. There was a large scale loss of human life and destruction of infrastructure during its course. Therefore, with peace came the emphasis on rebuilding infrastructure, especially roads, bridges, buildings and cities. The United Nations agencies also laid renewed emphasis on the concept of human dignity and the rights of an individual. All this helped in bringing the notion of development to the centre stage.

After the Second World War, the world became politically divided into the capitalist bloc lead by the USA and the Communist bloc lead by the USSR. These two competing worldviews and alternative development paths also helped to make the concept mainstream.

1.2.1 Defining Development

As stated earlier, the term Development relates to progress, advancement in stages, modernisation and economic growth. The term development is a contested one. According to Amartya Sen, Development is freedom. He further says, “freedom is both the primary objective to development and the principal means of development. The human being is the engine of change”. Is economic progress, for some, means well being of the individual. The overall development is growth, wellbeing, economic progress, freedom to pursue what we want to pursue development. From 1970s onward, development has come to mean social change. Questions related to the environment, sustainability and human rights are all part of the development discourse today. The Rights-Based Approach to development stressed the importance of the rights to basic needs – food, shelter, education, livelihood, among others.

1.2.2 Development Communication

Development communication makes strategic use of communication in the pursuit of development objectives. Development communication is positive, pragmatic

and practical and follows various approaches: Information, Education, Communication IEC approach, Behaviour change communication, advocacy, social mobilization and others. Development Communication includes social marketing and social advertising as well.

1.2.3 Development Journalism

Development journalism is a subset of development communication. It is centred on news related to development issues and does not include advertising and public relations. In India, development journalism is interpreted to be news and features relating to development issues. In certain countries, it is viewed as the information given by the Government.

Check Your Progress Exercise 1

Notes: 1) Use the space below for your answer.

2) Compare your answers with those given at the end of this Unit.

1) What is common between development communication and development journalism?

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.....

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2) What is development reporting?

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1.3 MAJOR DEVELOPMENTAL ISSUES AREAS

Many issues related to development came to the forefront after the Industrial Revolution in the 18th century. These stemmed from new modes of production, creation of capital and labour, rural to urban migration and transformation of society from traditional to modern. During the period after the Second World War, the development pathway was based on economic growth. The economists viewed the central issue of development as being characterized as the growth of the national economy. While the field of development journalism emerged in the 1960s, the issues covered have changed for a while.

1.3.1 The Economy

The economy has occupied the centre stage in development right from the Second World War onwards. The dominant paradigm of development that held sway

from 1945 to about 1970 considered development synonymous with economic growth. Development was defined in purely economic terms. It was a limited world view which considered economic growth to be an essential aspect of development. Economic growth was to be achieved by capital intensive investment in large projects. Development was to be measured in macroeconomic indicators such as Gross Domestic Product GDP and per capita income.

Economists such as W.W. Rostow had popular models of economic growth centred on mechanical processes of spurring savings and investment rates. **His proportion is related to modernisation theory.**

In the period immediately following the Second World War, development was considered an investment in infrastructure and rebuilding of many roads, bridges, dams, factories, buildings etc., which were destroyed in Western Europe.

Today, the economy continues to be central to our understanding of development, but the field's scope is much broader. Reporting on the economy involves the interpretation of statistics and trends to simplify and demystify complex issues. Often economic reporting is based on data. The economy broadly includes the primary, secondary, and tertiary or services sectors, with infrastructure and telecommunications being the most important. The primary sector includes agriculture, and the secondary includes industry and manufacturing. Sectors such as information technology, energy, transportation, financial and social services are considerable areas in themselves.

Journalists reporting on the economy must understand the Union and Budgets, Economic Survey, taxation policies and other policy announcements made by the Finance and Commerce Ministries. After economic liberalization in 1991, news stories reflect happenings in the stock, commodity and bullion markets, and monetary and credit policy announcements made by the Reserve Bank of India. The privatization, liberalization and deregulation that India witnessed post-1990 gave content for a variety of news. Taxation and corporate affairs are potential areas of economic reporting.

Economic indicators such as Gross Domestic Product GDP and Per capita income, indices of consumer and wholesale prices, industrial and agriculture production, employment, stock indices, money supply, trade balance, foreign currency reserves, the exchange rate of the Indian rupee are all material for economic and developmental stories. Comparative growth trends across regions or at different points of time all provide a good scope of such reporting.

The poverty rate or poverty line is a topic of debate and discussion in any democratic society. The state poverty data, national numbers and international comparisons are all subjects of good development stories. At the same time, poverty could be calculated by keeping in mind some minimum level of per capita income required to purchase basic amenities. More developed countries also factor in access to health and education. The World Bank releases data on global poverty, taking a base of earnings in terms of dollars per day. In India, the erstwhile Planning Commission used to release information on poverty trends in the country.

The emphasis on social aspects of development increased after the 1970s with the emergence of the participatory paradigm of development. The emphasis on industrialisation in the mainstream economic models of development was called into question. The concept of development was broadened to include social aspects apart from the economy. Health and education were two significant areas that gradually grew in importance. The Human Development Index (HDI) formulated by Amartya Sen, a composite index, measured social aspects of life beyond the Gross Domestic Product. It was an amalgam of the GDP growth rate, a measure of education, and health measures. Thus, it was broader than just a measure of wealth or the GDP growth rate. The HDI statistics are compiled and released annually in the form of reports published by the United Nations Development Programme. These are a good indicator of a country's overall development over a period and its comparison with other countries. The education sector is measured in terms of overall literacy rate and gross enrollment in schools and colleges. Health is generally understood in terms of the expected rate of longevity. This gives some indicator of the state of healthcare in a country, as well as nutrition.

There is an emphasis on including gender ratio in the context of development. For example, the access of girl child to education, nutrition for girls vis a vis boys, etc. There are also gender-adjusted income accounts and development statistics compiled by the UNDP as part of the Human Development Report.

Indian media and policy planning circles gave a lot of emphasis to analyzing changes in GDP figures. Unfortunately, the same emphasis is not given to analyzing HDI figures. India has not done well in HDI rankings in the region or other developing countries with similar economic growth rates. For a development journalist, the different aspects of a country's HDI can be a good matter for development stories since HDI incorporates social sector elements such as education and health.

Various government policies and schemes geared towards poverty removal and the lower-income strata of society are the subject matter of development stories. For example, the state-wise performance of the Mahatma Gandhi National Rural Employment Guarantee Act (MNREGA), Integrated Child Development Scheme (ICDS) and schemes for poverty removal or their impact on people at risk constitute some excellent areas of interest for a newsperson.

1.3.3 Government and Governance

Let us start differentiating Government and governance. Government is a structure, and governance is a process. The Government of any country is central to development issues. It is a vast news beat in itself as several ministries need to be covered by reporters. Government policy is yet another area for development journalists.

Governance has always been an important issue in development. Writing on social change and the social thinker Ibn Khaldun mentioned the quality of governance as one reason for the ascendance or decline of a civilization. The quality of governance or good governance has been emphasized in the World Bank literature from the 1980s onwards. Development stories based on the State as an actor or State as a reformer all delve into governance and policy planning.

Even economic stories based on trends on foreign direct investment are often related to the stability of the Government and reasonable governance measures. If the perception is of an unstable government or an ineffective or corrupt government, it isn't easy to attract foreign capital for investment. You will read more about it in the next Unit.

1.3.4 Environment

The dominant paradigm of development that prevailed from 1945 to the decade of 1970s was characterized by industrialisation, primarily through large scale projects. Governments pushed heavy capital intensive projects in infrastructure for development. Factories, steel plants, hydroelectric dams, roadways, bridges etc., were the road to modern life and development. Manufacturing was the keyword everywhere.

An unintended consequence of such industrialisation was pollution and environmental degradation. In Unit 2 on Environment Reporting, you have read that the environmental movement grew from a small beginning in the 1970s. Rachel Carson's book, *A Silent Spring*, highlighted concerns about the adverse impact of the dominant development models on the environment. Slowly public perception about the efficacy of heavy industrialisation as the road to development began to change. Some development reports of environmental damage and adverse impact on humans from pollution started surfacing in newspapers in developing countries. Alternative pathways to development using intermediate technology and small and medium scale projects started becoming popular after the 1970s.

The notion of sustainable development was discussed in various United Nations meetings and conferences. It helped to build public consciousness environment. Issues such as ozone layer depletion, carbon emissions, alternative energy and climate change, and pollution have become mainstream now. Alternative pathways to development that do not damage the environment are being explored. Today environment is an important beat. Various global environment accords are important stories. Environmental stories are interconnected with development, and many environmental stories are best understood by studying processes for a more extended period. Climate change and global warming are topics that provide the subject matter of many development stories. Alternative energy such as solar or wind has also emerged as a significant development and environmental reporting aspect.

The Indian news media is increasingly reporting on air pollution and its causes and its adverse impact on human health. The media has popularised the measures against air pollution. Issues such as particulate matter and crop burning etc., have become common knowledge because of sustained reporting by media.

1.3.5 Infrastructure

The dominant paradigm of development was centred on the importance of infrastructure and the economic growth rate. The infrastructure sector is substantial, with everything from highways to telecommunications being part of it. All these aspects are crucial for development. Although infrastructure is part of the economy, it is so vast a subject that it is generally studied and reported on as a subset – highways, waterways, railways, etc. In India, the railway's sector is so huge that the railways had their separate rail budget presented in Parliament

until recently. The railways are perhaps the most significant employer not only in India but the entire world. Development stories on Indian railways could cover passenger, freight, pricing, safety, customer service, new routes, track maintenance, employees issues etc. Telecommunications is another sector that has implications for the entire economy. Telecommunications systems act with a multiplier effect, impacting all the three sectors of the economy, primary, secondary and tertiary. Development stories concerning telecommunication could touch upon the spectrum, pricing, penetration of services, new applications and comparative growth trends across states or regions.

1.3.6 Agriculture/Rural

Agriculture is a large segment of the primary sector of our economy. Innovations in agricultural techniques formed the basis of the early work in development communication. Everett Rogers proposed his theory called Diffusion of Innovations based on agricultural innovations among farming communities during the 60s.

The agricultural sector is still vital to the economy. As a country develops, GDP accruing from agriculture generally goes down, while that of the services sector increases. The agriculture sector is a considerable employer in many developing countries. In India, productivity in agriculture is mainly low as the sector suffers from unemployment and underemployment. Some developing countries are heavily dependent on their agricultural sectors either as raw materials domestically or for export. Such countries have very little diversification in their economies and are heavily dependent on fluctuations in the price of agricultural commodities.

In India, the farming sector is crisis-ridden. Farmers often go bankrupt after taking loans which they are unable to repay. They often demand loan waivers, an increase in minimum support price and other benefits from the Government. In India, there is a lack of well developed cold storage networks. Thus a lot of agricultural produce rots and not processed. Spurt in farmer suicides is a stark reminder that the sector is bedevilled with problems. P. Sainath's book *Everybody Loves a Good Drought*, a compilation of his development reports on agricultural and development issues, throws light on the problems faced by the agriculture and rural sectors in India.

Apart from agriculture, rural issues include access to education and health. Livelihoods are also an important aspect. Various government schemes have targeted rural poverty and distress. There is generally a paucity of good news stories on rural issues. You will read more about it in Unit 7 of this Block.

1.4 REPORTING DEVELOPMENT

Some evergreen principles of reporting have stood the test of time, whatever the change in scenario.

1.4.1 Principles of Good Reporting

Good journalism is based on principles such as truth, objectivity, accuracy, fairness etc., and adherence to these principles differentiates a journalist from an activist. The job of a journalist is to tell the story, as it is, without taking sides. A journalist reports on the story, s/he is not part of the story. Development reporting also

adheres to these basic tenets. Good development journalism involves reporting on stories that are concerned with some aspect of social change. Reporting on development issues is not synonymous with propaganda on behalf of the Government or a non-Government organisation. Instead, it involves looking at developmental issues with a critical lens and separating the truth from the lies. It also means holding various stakeholders accountable.

Truth and Accuracy

Like any other area, reporting development issues also requires adherence to truth. Accuracy involves cross-checking facts to make sure that there are no errors.

Verification of facts is more important than ever before in the present time when digital media often indulges in the creation and dissemination of false and fake news.

Objectivity

In journalistic terms, objectivity implies covering the issue from all angles and perspectives to arrive at a 'balanced' view. For example, in a story about an industrial dispute and labour unrest, views from labour, management, government officials, and independent experts could all be considered. The journalistic notion of objectivity is different from the view of the social sciences that all enquiry is essentially subjective. The journalistic notion of objectivity is more functional in approach and a recipe for practising good journalism.

Fairness

Reporting should not be one-sided. If an allegation made the person or entity concerned, they should be allowed space to offer their opinion or rebuttal. All the relevant facts should be disclosed. Insinuations or partial disclosures which are motivated go against the tenets of good journalism.

1.4.2 Techniques of Good Reporting

Development reporting follows many of the principles and procedures of reporting in other areas. A news story follows a classic inverted pyramid approach. The most important information is given at the beginning, followed by less important information. What makes a subject newsworthy? Something has changed, it has an impact, and it is essential to the audience. For example, a new law or policy announced by the Government could be considered newsworthy. A disaster that affects human life is also newsworthy. Questions such as the magnitude of the change and impact thus determine the newsworthiness of the topic. Proximity is yet another determinant. For example, a tsunami affecting India directly is of more interest within India than a tsunami affecting Indonesia, even if the casualties are more.

Development features are written in a format different from a news story. Most are not required to follow the inverted pyramid format. A feature can adopt the narrative structure of a story with a beginning, a body and a conclusion. We have discussed in other units that there is no single way to write a news feature. However, good features typically personify an abstract issue and are best written from a human interest perspective.

Subject/Angle of the story

Generally unfolding current events trigger development news reportage. However, investigative stories require in-depth coverage and contextual information. The story's angle can come out of research - it can be based on current happenings that gain a particular prominence. The headline of the story is critical to grab attention in today's competitive media environment. It must relate to the subject matter and yet be catchy.

Reporting style

It should be audience-specific - the literacy levels, age, gender, cultural background, rural or urban and other parameters should be kept in mind. In reporting development, jargon should be avoided. When technical terms are used, these should be followed by a short explanation to connect with the reader/audience. The style should be simple, lucid and crisp. Most print and television stories are now shorter in length. Online stories can have more information as compared to print media stories.

Research-based

Development reporting should be research-based. Development issues are typically complex and have different dimensions and ramifications. Research in these areas is essential. Research can start with secondary material such as published books, journals, reports from independent and government agencies. It should be followed up with interviews with experts concerned, government officials, and persons affected by the issue in question.

Many development stories can be based on data, such as from the human development reports or Ministry reports on different economic aspects. Statistical data available from ministry reports/websites can be used for development stories pertaining to economic trends or health and education indicators etc. The UN agencies are often a good source of research material, from human development reports to health and climate change.

Visuals/ Infographics

Visuals add interest to a story and make it alive. A good photograph can literally speak volumes. A good photograph can personify or illustrate an abstract issue. Infographics and graphs can be used for illustrating complicated trends or statistics. Bar charts, pie charts, line graphs are among the more common ones. For online stories or television reportage, video clips can tell the story. However, a word of caution is needed. In this age of fake news, photographs can be photoshopped and morphed, leading to false news being created and disseminated. As with other forms of journalism, the source is essential in ascertaining the credibility of a news story.

Sources

The credibility of sources is key to the impact of a story. Sources quoted can be government officials, independent experts, people affected by the matter under consideration etc. The participatory paradigm ushered in from the 1970s gives more importance to dialogue, debate and discussion, which are participatory and democratic processes. Thus the angle of vision of the news or in-depth story has to be people-centric, rather than based on 'official sources of information. For example, quotes, photographs, audio or video clips should be affected by the

development processes. The development reporter should make all efforts to locate sources that will give insights into an issue that s/he is covering. As indicated earlier, research-based reports and publications can also be authentic sources.

Time frame

While news stories are based on recently unfolding events, development stories often need to consider processes to give context to the story. These may require research into the background of an issue. For example, a story on agricultural development in an area should cover how it is happening and who have been instrumental in bringing it about over a period.

Check Your Progress Exercise 2

Notes: 1) Use the space below for your answer.

2) Compare your answers with those given at the end of this Unit.

1) What are the principles of good reporting?

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2) What are the techniques of good reporting?

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1.5 REPORTING DEVELOPMENT IN DIFFERENT MEDIA

While some reporting principles on development remain standard across all media, some are more specific to a particular medium. In the following section, we will discuss reporting development issues in print, radio, television, film and new media. Historically, much of the development coverage has been related to disaster management or reportage of problems faced in developing countries. Positive development stories have had a much smaller share of the coverage.

1.5.1 Print

They are reporting development news stories like any other news story to follow the classic inverted pyramid style. Here the important question of who, what, when, why and how are all covered in the intro (first) paragraph of the story, with other elements of the story following in subsequent paragraphs. Most news stories are shorter than features, with the length of every story differing from

newspaper to newspaper. For example, The Hindu newspaper gives more excellent coverage and space to developmental stories than the Times of India.

1.5.2 Radio

Radio continues to be an effective medium for reporting on development as it has a comprehensive outreach and covers different strata of society. While radio sets may be declining, radios are being listened to more and more over mobile phones. Many public service announcements are aired repeatedly on the radio, as it is a suitable means of reaching the target audience. In India, we have the national broadcaster, All India Radio, and many private radio stations in operation. However, news and current affairs remain the purview of the state, as mandated by our Government. Private radio stations in India are primarily focused on entertainment. There is always scope for good radio documentaries and features on development. The role of radio is also essential in giving disaster warnings and coverage to disaster management.

One of the effective mediums for reporting on development is community radio. In countries such as Tanzania and Nepal, community radio is at the forefront of reporting development. Community radio programming is developed and aired by the community members themselves. It is genuinely participatory and reflects the voice and aspirations of people. In India, community radio is yet to take off in a big way. However, there is a sense of scepticism that the medium may be misused in some troubled regions in the country. Many licenses are yet to be allotted.

1.5.3 Television

Most of the private TV channels are based on advertisement-supported models. The exception is public service broadcasting. In India, most TV channels are striving to survive in a crowded, competitive market. The channels often adopt sensationalism and tabloidization to grab the attention of a fractured audience. In this scenario, the space for in-depth development stories is minimal.

1.5.4 Films

Documentary films are a good way to explore development issues with an in-depth approach. Documentary films are a vehicle to explore nuances of development which may not be available in mainstream TV news channels. Unfortunately, the channels showing development based documentary films are relatively few. Some channels in the infotainment sector, such as Discovery or National Geographic show documentary films.

Mainstream commercial cinema also has the capacity to create awareness about development issues. While the primary purpose of commercial cinema is entertainment and to earn profits, it can also be a vehicle for informing the public. For example the Hindi film *Taare Zameen Pe* created a lot of awareness about learning disabilities and dyslexia. The Hollywood film *Erin Brockovich* created awareness about industrial pollution and health hazards.

1.5.5 New Media

New media, including websites, e-publications and social media such as Facebook and Twitter, can be utilized effectively to report on development issues. Web

portals such as Scroll cover a range of development-related stories and have a wide audience. Since space is not a constraint in new media, detailed development stories can be covered in it. The new media is inherently multimedia, which allows audio and video clips to be inserted. Suitable links related to connected stories can be hyperlinked as well. Writing for social media differs with the medium. For example, Twitter is restricted in length, so generally, short opinions are aired. However, links to long stories can also be added. Facebook posts are generally to a general audience of 'friends'.

Check Your Progress Exercise 3

Notes: 1) Use the space below for your answer.

2) Compare your answers with those given at the end of this Unit.

- 1) What is the difference between reporting a development story for print media and for online media?

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- 2) How can films help in creating awareness about development issues?

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1.6 LET US SUM UP

In this Unit we discussed various issues related to Development studies. We started with the concept of development and then defined development communication and development journalism. Some major development areas such as economy, social sector, governance, environment, infrastructure and agriculture and rural development were also explained. It was observed that development issues are complex, interrelated, have a long time frame, and technical in nature.

We then moved on to describe the principles and techniques of development reporting. The principles of good journalism such as truth, accuracy, objectivity, fairness etc. hold for the development journalism as well, and the need for proper selection of format in development reporting was emphasized.

We also briefly looked at development reporting in different media – print, radio, television, films and new media and discussed the challenges in reporting development issues. Some pertain to lack of knowledge on the part of the journalist, while others relate to lack of authentic information. At times, problems

arise due to corruption or influences that seek to block stories and propagate false news. As a journalist, you need to stay vigilant and take due care in reporting development issues and give voice to the people effectively.

1.7 REFERENCES AND FURTHER READINGS

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Online Sources

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1.8 ANSWER TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress Exercise 1

- 1) All communication aspects, except for advertising and public relations approaches, are common to both development communication and development journalism.
- 2) Earlier reporting on development meant only economic growth. Since the 1970s, the arena of reporting on development has widened to include social issues, environment, gender, demography, nutrition, policy etc.

Check Your Progress Exercise 2

- 1) The principles of good reporting involve adherence to truth, accuracy, fairness, balance, objectivity etc.
- 2) The techniques of good reporting include research, use of appropriate reporting style, visuals, infographics, credible sources and angle of the story.

Check Your Progress Exercise 3

- 1) Print media stories tend to be shorter than stories on new media which are not limited by length. Stories on new media can be multi-media with audio and video links and can also be connected to related stories via hyperlinks.
- 2) Documentary films can explore development issues in-depth. Commercial cinema can use an entertainment format to create awareness about development issues. They have large audiences and large potential impact.



UNIT 2 GOVERNANCE ISSUES

Structure

- 2.0 Introduction
- 2.1 Learning Outcomes
- 2.2 Governance- Definition
- 2.3 Elements of Governance
 - 2.3.1 Transparency
 - 2.3.2 Participation
 - 2.3.3 Accountability
 - 2.3.4 Equity
- 2.4 Role of Mass Media in Governance
 - 2.4.1 Federalism in India
 - 2.4.2 Budgets
 - 2.4.3 Private and Corporate Sectors
- 2.5 Issues in Governance
 - 2.5.1 Procedural Lapse
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- 2.6 Reporting Issues Related to Governance
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 - 2.6.4 Some Guidelines for Reporting Governance
- 2.7 Let Us Sum Up
- 2.8 References and Further Readings
- 2.9 Answer to Check Your Progress Exercises

2.0 INTRODUCTION

In the previous Unit, we have discussed 'development' and its communication aspects. Governance is the system responsible for inclusive development, socio-economic and cultural progress and well-being of individuals and nations. Governments are structures that operate in specified legal and administrative frameworks responsible for the effective functioning of governance systems. 'Minimum government and maximum governance is the mantra which constantly appears in discourses to argue the case of 'governance for the transformation of people's lives for good'.

The strength of democracy is visible in pluralistic views, civil liberties and progressive political culture. All these factors have a direct correlation with the efficiency of governing bodies and structures. Thus it is imperative to have checks and public scrutiny of the systems of governance. Elections, parliamentary procedures and judicial scrutiny are some ways to achieve this objective. However, media, the Fourth Estate, has a pivotal role in safeguarding the principles of

democracy, good governance and accountability. Thus reporting anything related to governance by media is closely linked with the day-to-day life of all citizens of a country.

Reporting governance issues assumed greater significance in India after the economic liberalisation. The LPG (Liberalisation, Privatisation and Globalisation) policies were perceived to be catalytic forces in reducing the role of government in many sectors except defence and law and order. The policies held that the government should only be the facilitator and regulator and not the central actor in overall development. But, in practice, the absolute necessity of good and responsible governance is clear and visible in market-driven development activities.

As the scale and volume of development activities have increased considerably and the complex structures of development models have evolved through private participation, the role of media has grown beyond reporting the obvious. The development of journalism needs to be expanded to corporate companies, civil society, and global as well as domestic markets. It needs to reflect the growing disparities between the rich and the poor and is expected to play a central role in maintaining social equilibrium in a society where competition is the buzzword.

In this Unit we will focus on reporting governance as a specialised area. We shall look at the principles of journalism in the context of governance reporting and examine the issues and challenges in governance reporting in India. We shall also give some practical tips to achieve better results in governance reporting.

2.1 LEARNING OUTCOMES

After studying this Unit, you should be able to:

- define governance;
- describe the essential attributes of good governance;
- discuss the historical evolution of reporting governance; and
- explain reporting issues related to governance in the era of globalisation.

1.2 GOVERNANCE- DEFINITION

The State assumes the central role in governance. In fact, State is the apparatus for governance and it is the body that links different organs such as executive, judiciary, legislature, civil society, markets etc. The State is a set of institutions to administer, remove poverty, uphold social equality, empower weaker sections, control private players, guard national security and maintain law and order. Governance is the steering process of the State. Some of the definitions of governance are as follows:

‘The manner in which power is exercised in the management of a country’s economic and social resources for development’ (World Bank, 1992).

The World Bank identified four broad themes to capture the essence of this definition. These are Public Sector Management, Accountability, Legal framework for development and Transparency and information.

‘Governance can be seen as the exercise of economic, political and administrative authority to manage a country’s affairs at all levels’ (UNDP, 1997).

The UNDP explained good governance as :

- Participation of entire population in decision-making either directly or indirectly through elected representatives;
- Rule of law which establishes and protects human rights;
- Transparency which ensures free flow of information accessible to all;
- Responsiveness of institutions to stake-holders;
- Consensus orientation which ensures best interests of a society;
- Equity which guarantees equal opportunities for all;
- Efficiency which produces desired results;
- Accountability; and
- Strategic vision which gives a long term perspective.

“Governance is concerned with changes taking place in the organisation of the state, and with changes in its relationship with the private sector and civil society actors” (Mathur, 2009)

Good governance is always aimed at bringing positive change. Now the word ‘development’ itself has been contested and rephrased to bring in the concepts of equity, inclusion and assimilation of ethnicities and culture. Thus the word ‘social change’ has gained currency in recent development discourses. Mathur correlates the efforts of the state, private sector and civil society in changing a society through efficient, entrepreneurial and transparent governance.

The conduct of governance in India has been dealt in detail in every form of mass media since the independence. The slow pace at which the Indian economy moved till 1990’s has been replaced with several institutions of governance, legal and administrative frameworks, global integration, international models of businesses, private participation and civil society participation. The accelerated pace of events transformed the lives of people and there is a growing urge among public to have better service delivery and desire to participate in matters of governance. The Cabinet resolution (2015) which set up the Niti Aayog in place of the Planning Commission says”, ‘Government and governance have to be conducted in an environment of total transparency—using technology to reduce capacity and thereby the potential for misadventures in governing’.

Although such administrative reforms are essential for efficient governance, the watch-dog role of mass media and best practices of development journalism are foundational to ensure transparency and accountability. This has been evident in many aspects such as sensitisation of people, follow-up of steps taken by institutions, keeping track of unhealthy practices such as corruption and nepotism, lack of transparency, loss to the state exchequer, deviation from administrative procedures, violation of law, socio-economic impact, displacement and marginalisation.

Good governance leads to an overall development of a country and its population. It brings about positive changes in the lives of the people. It should be inclusive in nature and the results are expected to be distributed equally. It is very difficult

to achieve this goal and several cases of unequal distribution and marginalisation surface with every development initiative. To control this, we need to understand the real meaning of ‘good governance’ and the parameters for measuring good governance.

Activity-1

Identify a Government of India flagship programme (Swachh Bharat/ MGNREGA/ Ayushman Bharat) and find out how it is being implemented by various agencies of the government. List out the issues published in mass media during implementation of such schemes.

2.3 ELEMENTS OF GOVERNANCE

From the definitions given above you would have seen that several elements constitute the process of governance. The term governance has been used in a broad sense in these definitions. However, in a modern democratic society governance should have specific meaning to make it to ‘good governance’, a term which gained currency in recent discourses. Good governance is essential for sustainable development.

According to the United Nations General Assembly World Summit (2005) sound economic policies, strong democratic institutions responsive to the needs of people and improved infrastructure are the basis for sustained economic growth, poverty eradication and employment generation. Freedom, peace and security, domestic stability, respect for human rights, the rule of law, gender equality and market-oriented policies are essential and mutually reinforcing to establish the system for good governance. Commitment to ensure justice for all and protecting democratic principles are paramount in good governance.

The above mentioned attributes of good governance make the role of mass media very important in achieving the desired results. Here, people are at the centre and all other stake-holders of governance are expected to take steps primarily in the interests of people at large. Hence an external mechanism such as mass media is required to look at the process of governance in an impartial manner. How this can be done? Let us address this question by analysing the role of mass media in ensuring the essential components of good governance.

2.3.1 Transparency

Transparency in governance means the policy decisions taken by the government which have an impact on citizens lives should be in public domain and people should have access to that information. All decisions taken by the government should follow existing norms and legal framework. Merely publishing the information will not ensure transparency. If citizens have doubts on the subject-matter, the government is expected clear these doubts and if needed, change the decision.

The Right to Information Act implemented in 2005 is a right step towards bringing greater transparency in governance. Please refer unit 15 for detailed information. Through public disclosure policies international agencies reveal details of contracts and various aspects of governance processes.

The role of mass media is to ensure transparency in all aspects of governance by constantly enquiring and verifying the information given by various agencies. Media is also expected to interpret information in an understandable and non-technical manner for its audience. Media should make use of the RTI, Whistle Blower policy, NGO interventions and public disclosures to educate its audience on steps taken by various agencies as part of governance.

Although there are institutional mechanisms to ensure transparency in governance it has been observed that governance is not transparent in the letter and spirit. Mass media should look for the gaps in information, interpret it with a common man's perspective and present it in an effective manner. There are powerful people who try to thwart the true spirit behind the RTI Act and withhold information. Journalists who are engaged in reporting governance face such people and their attempts to kill the spirit of good governance.

2.3.2 Participation

Good governance can be achieved only through participation of people. Those who are affected by government decisions should be consulted in advance and adequate compensation should be paid to them when required. The participatory approach in governance may reduce conflicts between the state and the public. All government actions should be initiated after due consultation with people's representatives. However, we have seen several instances of lack of communication or exclusion of stake-holders in implementation of development programmes by governments which led to conflicts between the state and affected people. These incidents naturally get media headlines.

Journalists should look for gaps in participatory approach and report the root cause of conflicts. The socio-economic impact assessment is an important step in gathering information about people who are going to be directly affected by any development programme. In order to build trust and confidence among people, mass media is expected to publish details of the project which were not highlighted earlier by authorities. Awareness of the project in an early stage will empower people to participate in the decision-making process of the project. Thus mass media can give voice to the people in an effective manner.

Social media has emerged as a powerful means to voice people's opinion on any issues of governance. The absence of gate-keeper in social media has two aspects. While empowering people, social media can also be used as a platform to spread half-baked truths and propagandist material. Since majority of the social media messages are highly polemic in nature it is very difficult to have an objective understanding of issues regarding governance. In this context, the role of legacy media has become more significant in providing a comprehensive view of governance.

2.3.3 Accountability

In a democratic system of governance, accountability lies both with political as well as bureaucratic functionaries. Both are answerable to the people for their decisions and actions. Being accountable to people requires responsiveness to the needs and issues of the people. Understanding 'public interest' through transparent interactions and a participatory approach can help the government be accountable. But we have seen contradictions in defining the term 'public

interest' on many occasions. For example, a big chemical industrial unit proposal may be of 'public interest' to those who gain employment and reap economic benefits from it. It can also be seen as a symbol of national development, and the government may claim the credit. But it will not be a good experience for the people who will face environmental pollution, displacement and other problems. The government is supposed to be accountable to all sections of society.

2.3.4 Equity

Ensuring equal opportunities for all is extremely important in the process of good governance. Good governance has eight major characteristics. It is participatory, consensus-oriented, and accountable. Transparent, responsive, effective and efficient, equitable and inclusive and follows the rule of law. It ensures that corruption is minimized, the views of minorities are taken into account and that the voices of the most vulnerable in society are heard in decision making. It is also responsive to the present and future needs of society. (United Nations Economic and Social Commission of Asia and the Pacific (www.unesco.org)).

Opportunities are to be given to stakeholders of the programme or project and the lower strata of society, who are deprived of better living. Thus good governance should look at the initiatives or programmes of the government from the perspective of concerned people. For example, a port construction proposal should not overlook the concerns of fishers living in the area. The coverage should give a holistic picture that reflects the hopes and aspirations of all people concerned. It should throw light on the positive and negative impact of such initiatives on the lives of people. Journalists have a moral responsibility to highlight problems and the need for corrective measures.

Apart from the broad areas of good governance, as a journalist, you should also look at the prevalence of the rule of law. It would be best to look at the system's overall efficiency in completing the targets on time and building consensus for the public good. Your coverage should be futuristic, which should help the authorities fine-tune strategic vision for generations to come.

Check Your Progress Exercise 1

Notes: 1) Use the space below for your answer.

2) Compare your answers with those given at the end of this Unit.

1) Define Governance.

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2) What is good governance?

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3) What is participatory approach in governance?

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2.4 THE ROLE OF MASS MEDIA IN GOVERNANCE

Governance in a democratic system broadly functions at three levels-local, state and national. With the advent of globalisation, international affairs cast significant influence on all these levels of governance. The local governance in India has institutional frame-work of elected representatives and has economic authority to carry out local development. The federal nature of governance is clearly defined in the Constitution. Globalisation paved the way for market forces and civil society into the realm of governance.

Mass media intervention is significant at all three levels. The national scenario gives an opportunity to media to have a look at the overall progress of society against the promised agenda of governance. It also highlights regional imbalances and income and gender asymmetries in growth and progress. Mass media coverage of governance at national level compares macro-economic scenario with social reality at the grassroots. At the state level, media coverage of governance looks at the allocation, distribution and utilisation of resources. The local governments have significant role in transforming the lives of people with development initiatives. However, media should be aware of the legal, economic and constitutional framework that establishes linkages between these arms of governance.

2.4.1 Federalism in India

Our Constitution outlines the executive, legislative and financial powers of the centre and states and lists out the subjects that fall under them. The subjects under the Union List, State List and Concurrent List clearly demarcate the legislative powers of the Centre and the States. However, in matters of conflict of interests, the Central Act will prevail and the Union Government has overwhelming power to legislate over the State. A portion of the taxes and duties is given to the states through the statutory authority namely, the Finance Commission appointed by the President every five years. Although the terms of reference for fund devolution are prepared after consultation with stake-holders, contentious issues still remain.

Till 2014, the Planning Commission set out the parameters for development programmes to be implemented in the states. The parameters were discussed in the National Development Council (NDC) chaired by the Prime Minister, Chief Ministers presented the cases of their states. All state plans (specific governmental development programmes in all sectors including basic services which were to be implemented in a financial year) had to be approved by the Planning Commission and the share had to be fixed in advance.

In 2015, the Planning Commission was replaced by Niti Aayog (National Institution for Transforming India). While Planning Commission had powers to evaluate, allocate and approve state plans, Niti Aayog does not enjoy these privileges and such powers now rest with the Ministry of Finance. It is more of an advisory body which concentrates on 'strategies' of governance. The Planning commission was more focused on the process of governance. Niti Aayog is aiming at nurturing cooperative federalism which ensures the synergised functioning of national, state and local governments together.

Local governance is a system to empower grassroots level democratic institutions such as panchayats with administrative and financial resources. After the 73rd and 74th amendments to the Constitution in 1993, panchayati raj institutions have been given autonomy to formulate their own development strategies with greater participation of people at the local level.

Mass media has to look at these arms of governance to highlight the issues that may emerge in regard to cooperative federalism. Political compulsions, regional disparities and scarce resources always create problems. A reporter should be equipped with the basic knowledge of how these constitutional bodies function to bring out the x-factor in their news stories.

2.4.2 Budgets

Budgets are annual financial statements of national and state governments. These documents are unveiling policy, providing macro-economic scenario, listing the achievements and challenges, placing the priorities of governance and operating in many other areas relevant to people. You should look at the budget in terms of its outcome and impact and give an objective picture where the bodies of governance fail to deliver the promises made in a budget. While doing so, you should be able to reveal disparities in allocation, miscalculation of priorities and highlight the demands of neglected sections of society.

2.4.3 Private and Corporate Sectors

In economically liberalized India, governance does not confine to elected representatives of people and bureaucracy. It is also concerned with the role of private and corporate sectors who handle a major chunk of the GDP. It is important for you as a governance journalist to understand the nuances of private capital, investments and corporate business and its linkages with the official arms of governance. Corporate affairs and policies which aim at investment promotion are essential components in employment generation.

Hence, you should trace the flow of foreign and domestic capital and factors which influence trade and commerce. The administrative mechanism and legal and institutional frameworks of capital markets are also subjects that fall under governance. Thus it can be argued that all the areas which touch upon the lives of people have governance angle in it.

2.5 ISSUES IN GOVERNANCE

Some important issues affecting governance pertain to procedural lapses, corruption, lack of environmental concerns, displacement and marginalisation and infrastructure. Let us look at them in detail.

2.5.1 Procedural Lapses

All bi-party agreements in governance operate according to the stipulated conditions and are framed within the legal system of India. Any deviation needs to be examined and reported. Stringent procedures are laid down for any programme of governance to plug leakage of public money. Procedural lapses or efforts to dilute procedures in governance lead to corruption. Hence you should first study the legal and procedural framework, analyse it and look for lapses. Such lapses always make good headlines.

2.5.2 Corruption

India ranks high in corruption indices prepared by international agencies. Corruption denies rightful benefits to citizens. Any story which is done on corruption in governance is a service to the nation. Doing stories of corruption requires an in-depth study, use of all information avenues, contacts in official systems, help of experts and a thorough enquiry into the matter. This requires investigation skills.

2.5.3 Lack of Environmental Concerns

Modern democratic governance is aimed at establishing systems of sustainable development with environment as its main concern. Hence any development programme has to be sustainable and environmentally suitable. The underlying presumption in any development initiative is effective utilisation of natural resources. This principle should be at the core of governance reporting. A 'green eye' is absolutely necessary while looking at the feasibility of a development programme. There are institutional mechanisms such as green tribunals, environmental impact assessments, and compensatory afforestation to safeguard environment while implementing development projects. It is the duty of the bodies of governance to ensure the effective functioning of such mechanisms. Media has an all important role in keeping vigil in this regard.

2.5.4 Displacement and Marginalization

If displacement and marginalization of people keeps on increasing, it is fair to assume that effective governance is lacking. Building infrastructure, heavy industries and manufacturing units lead to displacement of people and in turn to resistance and conflicts. The role of mass media is to highlight the plight of displaced people and seek remedial measures. The displaced people should not become victims of development, instead be participants and stake holders in development. The rehabilitation, resettlement, compensation and livelihood support for them should be debated in mass media platforms.

2.5.5 Issues Concerning Infrastructures

Infrastructure is the basic foundation on which the factors of production meet to produce output. It refers to large-scale public systems, services and facilities that are necessary for an economic activity. In the era of economic reforms one of the major roles of government is to put in place effective and sufficient infrastructure which acts as a catalyst in economic growth. With the influx of foreign and domestic private capital, the government's role in the building of infrastructure mainly concentrates on establishing atmosphere for investments.

Public Private Participation (PPP) is being projected as a viable method to kick start infrastructure projects and to streamline investments, technology, labour and skills. BOT (Build, Operate, Transfer), BOOT (Build, Own, Operate, Transfer), BOLT (Build, Operate, Lease, Transfer) etc are some of the options given by the Government of India to materialize PPP projects. But we have seen many PPP projects which lag behind the time schedule. Some reasons could be corruption, contract raj, lobbying to maximize the profits by private players and bureaucratic lethargy. The projects also cause displacement of people and levying of user charges. In most cases transparency and accountability is missing and so are the measures for the affected people. All these gaps need to be looked into and problems highlighted.

Check Your Progress Exercise 2

Notes: 1) Use the space below for your answer.

2) Compare your answers with those given at the end of this Unit.

1) Identify some issues affecting governance.

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2) What are the issues concerning infrastructure?

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2.6 REPORTING ISSUES RELATED TO GOVERNANCE

We have seen that governance is closely related to development and well-being of the people. But development stories without sensational angles such as political maneuvering, inaction, corruption etc. gets poor coverage. For example, the tragedy of farmers' suicide often does not get media space it deserves. Only when it becomes a political issue with high profile visits to rural India, its news value increases. Similarly, if an NGO or government department does a good job in promoting vaccination among tribes it hardly finds space in media. But if a child develops health problems due to vaccination we can expect a big coverage. So development news needs a balanced approach. It is unfortunate that bad news becomes the news of the day while we talk of governance and development in India. Here we should look at the theoretical construct of news to develop right perspective on governance and its operation.

2.6.1 Elements of News

You are aware of many definitions of the term 'News'. In practice, we see that most of the news reports look at the gray side of life. Gatlung and Ruge (1965) enumerate a list of factors that determine news value.

- **Threshold-** A big story that has an extreme effect on a large number of people.
- **Frequency-** Events that occur suddenly and fit well with an organisation's work schedule. Sometimes events that occur at the midnight may not get covered.
- **Negativity-** Bad news is more exciting than good news.
- **Personalisation-** Events that can be portrayed as the action of individuals will be more 'attractive'.

By and large, Indian media coverage fits well with these parameters in the area of governance. A governmental action which may have a positive impact on public life does not enter the big news cycle. For example, if a tsunami warning system is installed it may find a single column space in the inside pages of a newspaper or two lines in 'News in Brief' (NiB) of a TV news bulletin. But if it fails to deliver the desired results the story will jump to the front page and make TV Headlines. Then the media will remind us of the potential impact of a second tsunami, the frequency with which tsunami hits the coasts of Indonesia and Japan, gray reminders of the earlier tsunami disaster and the people responsible for this failure. This is justified in one sense. Proper governmental action is pre-defined and expected. So if it carries out the expected tasks there is nothing out of ordinary. But contrary to expectations, if something happens, it becomes news and media reports it in a critical way. However, media will fail in its duty if it does not give adequate information on the governmental action well in advance. If the information has not reached people they will not benefit from it.

We are familiar with innumerable cases of corruption and irregularities in government programmes. Quite often, the implementation period of a project may stretch to double the time envisaged. Sometimes operational problems such as lack of funds, land allocation, expertises etc. also retard the progress of projects undertaken. If reporters are ready to look into the issue of delay, that will make headlines. Obviously, many facts are concealed by the authorities that affect proper reporting of governance.

Activity-2

Identify any development project carried out by the local self government (LSG) in your residential area. Find out the issues on its implementation. Prepare a news report through the lens of good governance practices.

2.6.2 Rethinking 5 W's and 1 H

You are aware of the classic lessons of reporting given by Rudyard Kipling which start with asking the six questions, 5W's and 1 H. While reporting on governance, the questions Why and How dominate over others. Other four questions (Who, What, When and Where) are answered by the responsible persons well in advance in the public domain. Since we have the RTI, which is characterised by an active and responsive civil society, no government can withhold direct and obvious information. But it may be evasive on two crucial questions - Why and How.

The rationale behind certain policy decisions is always questionable though the government enjoys legal backing for its decisions. Similarly, the modus operandi through which a government decision is implemented may be beneficial for the actors involved. So if you want to strike the right note in journalism of governance, you need to be equipped to ask the right questions to find the answers for 'Why' and 'How'. In such cases constitutional institutions such as the Comptroller and Auditor General of India, various courts, quasi-judicial organisations such as Lok Ayukta, Ombudsman etc. have a crucial role in bringing out the truth behind the failures of governance. You should keep track of such institutions and their activities.

2.6.3 Bureaucratic Affinity

Reporters tend to stick to the bureaucracy. Beat reporting actually confines journalism to a fixed perimeter and the events that happen outside this perimeter may get less attention by the media at the initial stage. This is particularly important in governance reporting. According to Mark Fishman, reporters are guided by the 'principle of bureaucratic affinity'. He perceives media houses as a form of bureaucracy with a rigid hierarchy. He says, 'Only other bureaucracies can satisfy the input needs of a news bureaucracy'.

But if journalists who report on governance decide among themselves to perform the role of 'watchdog' instead of 'lapdog', genuine sources can be cultivated among the bureaucracies. The example of Watergate scandal can be cited here. When the US President Richard Nixon manipulated the investigation agencies of the country to peep into the actions of the Democrats, Washington Post reporters Carl Bernstein and Bob Woodward were able to reveal the presidential conspiracy in 1972. But the unknown whistleblower known as 'deep throat' was the real igniting force behind the whole episode. Only in 2005, the identity of the 'deep throat' was revealed to the public as none other than FBI Asst. Director, Mark Felt. Although we quote this event to explain the roots of investigative journalism, this gives us an insight into the importance of cultivating genuine sources in bureaucracy to report governance effectively.

2.6.4 Some Guidelines for Reporting Governance

The importance of transparency in governance is being increasingly realised. This realisation has come from inefficient implementation of projects, corruption, huge wastage of public money and large scale marginalisation of the poor. The Right to Information Act, The Right to Services Act, whistleblower policy (protection to persons who point out violation of norms and corrupt practices) are some mechanisms to ensure transparency which should be exploited by mediapersons to inform people.

The following are some tips for reporting governance issues:

- A careful study and observation of society and people to build a perspective on its needs and aspirations, income divide, resources, political character, demographic profile, gender equality, problems after urbanisation, loss of livelihood and land etc. are absolutely necessary for a journalist.
- The ability to analyse various aspects of government programmes and their implications for all sections of society is the next step.

- The observations of society and analysis of governmental intervention in that society should be synthesised with a view to lead to betterment of people by reducing all forms of inequalities.
- While writing or packaging a story you should keep in mind that the technical details of the programme are translated in terms of its impact. In other words, too many specifics not known to the common man will be a distraction.
- If your ‘Why’ and ‘How’ make the authorities answerable, your story will hit the target. Else, it will be forgotten very soon.
- Beware of ‘bureaucratic affinity’ and tendency to stick to official version. Be skeptical. Handle any press releases after gathering background knowledge and literature.
- Reporting announcements of ministers promoting big development programmes alone will not touch the core of governance journalism.
- Cultivate genuine sources that can blow the whistle. Exposing corruption in the implementation of development programmes is a powerful form of governance journalism.
- Be a researcher always. Keep track of markets, private players, and civil society.
- Marginalisation of a weaker minority after big development programmes and fall in prices of agricultural and traditional commodities is a regular feature in globalised societies. Safeguarding them through carefully articulated arguments and showing the pitfalls in governance is a matter of great importance.

Check Your Progress Exercise 3

Notes: 1) Use the space below for your answer.

2) Compare your answers with those given at the end of this Unit.

1) What are the elements of news according to Galtung and Ruge?

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2) What is the significance of ‘Why’ and ‘How’ in reporting governance?

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2.7 LET US SUM UP

Reporting on governance issues has assumed greater significance since globalization. The art of governance reporting is to be learnt by acquiring a comprehensive knowledge on how governance is being carried out in a democratic society, the role of the Union government, the States, the Local Self Governments and the markets. We have seen the absolute necessity of good governance in a market-driven society to establish safety nets and enhance trickle-down effect.

Good governance demands a vibrant media which can pinpoint lapses in governance in a democratic society. The classic lessons of reporting will not be sufficient to make a mark in governance reporting. Greater thrust should be placed on questions such as 'Why' and 'How', bureaucratic affinity should be channelized to yield results. In addition, gaining expertise in the use of RTI and whistle blower policy are some of the requirements for a journalist who covers governance. A researcher's mind is best suited to a governance journalist. The emergence of such media will take place only if the world-view and experience of 'the common man' gets truly reflected.

2.8 REFERENCES AND FURTHER READINGS

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2.9 ANSWER TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress Exercise 1

- 1) Governance is the system which is responsible for inclusive development, socio-economic and cultural progress and well-being of individuals and nations. Governments are structures which operate in specified legal and administrative frameworks responsible for the effective functioning of systems of governance.
- 2) Good governance includes freedom, peace and security, domestic stability, respect for human rights, the rule of law, gender equality and market-oriented policies. Commitment to ensure justice for all and protecting democratic principles are paramount in good governance.

- 3) Participatory approach to governance involves giving voice to all concerned with a development programme giving space for free expressions, and empowering people to participate in decision making process of the project.

Check Your Progress Exercise 2

- 1) Some issues in governance are procedural lapses, corruption, lack of environmental concerns, displacement and marginalisation, among others.
- 2) The issues concerning infrastructure are failing business models, contract raj, resistance to cost for services, displacement, and bureaucratic lethargy, among others.

Check Your Progress Exercise 3

- 1) According to Galtung and Ruge, the elements of news are threshold, frequency, negativity, and personalisation.
- 2) Raising the questions - why and how enable the journalist to go beneath the surface of story and provide in depth answers.



UNIT 3 AGRICULTURE AND RURAL ISSUES

Structure

- 3.0 Introduction
- 3.1 Learning Outcomes
- 3.2 What is Agriculture and Rural reporting?
 - 3.2.1 Agriculture and Rural Reporting: An Overview
 - 3.2.2 Significance
 - 3.2.3 Agriculture and Rural Reporting in India
- 3.3 Issue in Agriculture and Rural reporting
 - 3.3.1 Agriculture Technology
 - 3.3.2 Farm distress
 - 3.3.3 Poverty
 - 3.3.4 Marketing agriculture commodities
 - 3.3.5 Government Policies
- 3.4 Media and Agriculture and Rural Reporting
 - 3.4.1 Dissemination of farm technology
 - 3.4.2 Selection of Media
- 3.5 Writing for Agriculture and Rural Communication
- 3.6 Let Us Sum Up
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- 3.8 Answer to Check Your Progress Exercises

3.0 INTRODUCTION

Agriculture is the mainstay of the Indian economy. With an agriculture production of \$375.61 billion, India is the second-largest producer of agriculture products. India accounts for 7.39 per cent of total global agricultural output. The contribution of agriculture to India's economy is much higher than the world's average (6.4%). The contribution of agriculture to the GDP is 15.4%. However, India has been witnessing farm distress for the past few decades. There are untold stories and unheard issues which need to be documented. Rural and agriculture reporting is a beat that is essential for developing countries like India. Keeping this in view, in this Unit, we shall discuss the significance of agriculture and rural reporting. We shall examine the history of the agriculture/rural beat within the Indian context. We will also discuss writing and production skills required for agriculture and rural reporting.

3.1 LEARNING OUTCOMES

After studying this Unit, you should be able to

- appreciate the significance of rural and agriculture reporting;
- discuss issues related to rural and agriculture reporting; and
- develop techniques in agriculture and rural reporting.

3.2 WHAT IS AGRICULTURE AND RURAL REPORTING?

In media, the agriculture sector is considered as a part of rural reporting. Since agriculture is the mainstay of rural livelihood, we generally tend to discuss each other. Therefore, for brevity and convenience, we will use both the terms agriculture and rural interchangeably. However, our focus will be on agriculture reporting.

Agriculture journalism refers to the reporting of the processes of food production and distribution. It essentially deals with the political economy of the agriculture sector. It discusses the marketing strategies and available marketing options. Agriculture journalism targets farmers by informing them about the best agriculture methods and the latest technical innovations. Agriculture journalism beat essentially functions as the link between governments, farmers and readers.

Agriculture journalism's importance grew manifold as technical developments in agriculture began to take place. New farming technologies, chemical manure and pesticides, hybrid seeds, new storage system etc., necessitated that information gets disseminated far and wide. Agricultural journalism is widely perceived as positive journalism, which is beneficial for the farmers.

3.2.1 Agriculture and Rural Reporting - An Overview

There are varied perspectives about the beginning of agriculture journalism. It is widely believed that the development of technology in agriculture was an outcome of the Industrial Revolution. It is essential to know that at the time of the Industrial Revolution, agriculture was the mainstay of the economy across the globe. Hence, the impact of the Industrial Revolution on agriculture was quite profound. During this period, massive resources were invested in research on 'controlling nature and increasing agricultural yield. Around the mid-1880s, farm journals started appearing across the countries. The advent of the printing press paved the way for the publication of farm journals.

In India, Girish Chandra Bose edited the English language Agricultural Gazette, first published in Kolkata in 1885. In 1865, Sir Syed Ahmad Khan, the founder of Aligarh Muslim University, asked J.H. Princip, the Collector of Aligarh, to translate (from English to Hindi) three books on weather and its relationship to agriculture, agriculture methods in Europe and agriculture equipment. He also established a science society that published a weekly newspaper, The Aligarh Institute Gazette. The Royal Commission on Agriculture recommended the establishment of the Imperial Council of Agriculture Research in 1929. It was mandated that it would "act as a clearinghouse of information" on agriculture. In fact, before 1906, Imperial Agriculture Research Institute at Pusa, Bihar, had already started publishing journals, books etc., on agriculture. A donation of £30,000 from an American millionaire, Henry Phipps, helped establish the Imperial Agriculture Research Institute in Darbhanga district of Bihar. A section of people claims that the village became known as PUSA, an acronym, 'Phipps of USA'. The institute was shifted to Delhi on 29th July 1936 after a devastating earthquake on 15th January 1934. After Independence, the institute has been renamed as Indian Agricultural Research Institute (IARI). The name Pusa, however, has stuck to the institute.

In the 1960s, India initiated the Green Revolution in a bid to increase self-sufficiency in food production. As per the US Library of Congress website:

The introduction of high-yielding varieties of seeds after 1965 and the increased use of fertilisers and irrigation are known collectively as the Green Revolution, which provided the increase in production needed to make India self-sufficient in food grains. The programme was started with the help of the United States-based Rockefeller Foundation and was based on high-yielding varieties of wheat, rice, and other grains that had been developed in Mexico and the Philippines. Of the high-yielding seeds, wheat produced the best results. Production of coarse grains—the staple diet of the poor—and pulses—the primary source of protein—lagged, resulting in reduced per capita availability.

The total area under the high-yielding varieties programme was a negligible 1.9 million hectares in 1960. Since then, growth has been spectacular, increasing to nearly 15.4 million hectares by the financial year 1970, 43.1 million hectares by the financial year 1980, and 63.9 million hectares by 1991.

The Government of India went all out to promote this initiative. As a result, agriculture communication became more formalised and structured. The White Revolution provided another boost to agricultural communication. The National Dairy Development Board initiated it in the 1970s to promote self-sufficiency in dairy production.

The Government of India undertook a project of agriculture communication for the dissemination of information. However, the real challenge for the agriculture journalists was to report about the problems and sufferings faced by farmers.

3.2.2 Significance

Indian economy is mainly agriculture-oriented. According to the Economic Survey Report 2017-18, agriculture employs 50% of the country's workforce. The ancillary sector of agriculture is also reasonably large. With the proliferation of modern farm technology, people need to be increasingly informed about the latest technological innovations to get better productivity. Also, the farmers need to be apprised of government schemes to help them. Thus agriculture communication can empower the agricultural and rural communities in a big way. A journalist reporting on agriculture and rural issues should feel committed towards these communities to ensure that the voice of the voiceless is heard.

3.2.3 Agriculture and Rural Reporting in India

We have discussed the history of agriculture reporting in India, and the two specific agriculture and rural initiatives, namely, the Green and White Revolutions, which led to the communication revolution. It will be helpful to discuss these initiatives in detail.

Green Revolution

The green revolution was essentially about the bio-engineered seeds, generally called HYV or High Yielding Variety. This works in combination with chemical fertilisers, pesticides and heavy irrigation. The project was primarily implemented in Punjab, Haryana and Uttar Pradesh. However, Punjab remained the core state.

Punjab was known for having a massive dry land area. Its land was extensively irrigated by the canals constructed by the British, which turned the fortunes of Punjab. It produced a rich farming ecosystem, and prosperous farmers were the potential adopters of the new technology.

Radio was successfully used during the Family Planning programme launched in the early 1960s by the Government of India. It was also used quite successfully in the United States of America to disseminate information on agriculture technology. This motivated the Government of India to experiment with the help of All India Radio. The Green Revolution was essentially a part of an international initiative towards increasing food production. This initiative was spearheaded by the Rockefeller Foundation and the Ford Foundation. The international success story of communication in the agriculture sector inspired India to formulate a communication strategy for agriculture. Accordingly, in 1966, the Farm and Home Units were established at seven centres of All India Radio.

In the wake of the Green Revolution, many agriculture universities mushroomed across the country. The first initiative towards agriculture education was in 1936 when Madras launched a four-year bachelor's degree programme. At the time of Independence, there were as many as 17 colleges offering engineering courses in agriculture. Post-Independence, agriculture education came to be recognised as a major national priority to help the country achieve food security. The first agricultural University was set up at Pantnagar in 1960, which paved the way for establishing other agricultural universities across India. The USAID and American Land Grant Universities were instrumental in helping the state agriculture universities in India.

Extension Education was one of the salient features of agriculture universities. Extension Education refers to the process of teaching rural people how to live better by learning ways to improve their farm, home and community institutions (Leagans, 1961). Extension Officers, who worked as foot soldiers of the Green Revolution, were engaged by the Farm and Home Units of All India Radio to communicate with the farmers directly. Farmers would assemble at a particular place to listen to the daily broadcast, write down their questions on a postcard and post it the next day. They would get the answers to their queries within a week. Successful farmers would then be interviewed. The Rural Radio Farm Forums were modelled on a highly successful project in Canada. The Indian version of the project was a joint venture of the Ministry of Information and Broadcasting and UNESCO. This synergy between the Farm and Home Units of AIR and agriculture universities ensured smooth dissemination of information regarding the Green Revolution. Such was the impact of the programmes that fertiliser was named Radio Khad and a particular variety of paddy was named Radio Rice.

Krishi Darshan

The television wing of AIR launched a programme for the farmers called Krishi Darshan on January 26, 1966. It was initially launched as an experimental project in 80 villages around the then Union Territory of Delhi. Television sets were distributed, and community viewing was encouraged. Therefore, Krishi Darshan was more of a community experience rather than merely an agriculture communication programme. Prominent technocrat Kiran Karnik's observations in this regard are pretty pertinent. He states, "[it] had social ramifications because

of the 'community set' installed in each village, resulting in a congregation of viewers that cut across caste, class, gender and age. An independent evaluation proved the impact that TV programmes could have and strengthened the case for expansion of TV. From a luxury that a developing country could ill-afford, TV got recognition as an investment that could contribute to economic growth and social change. Krishi Darshan, much-reviled in later years as the epitome of "boring" programmes (especially by the growing middle-class urban elite), was thus the genesis of developmental TV in India".

SITE

The Satellite Instructional Television Experiment (SITE) was conceived in the mid-1960s after Krishi Darshan became an extremely successful programme. With this, it was proved that mass media can reach out to the people living in the remotest corners of the country. SITE was a collaborative effort of NASA, Ford Aerospace, General Electric, Hughes Aircraft and Massachusetts Institute of Technology under the aegis of then-nascent ISRO. This initiative was essentially targeted at generating development communication specifically meant for 2400 sites across the country in Andhra Pradesh, Bihar, Karnataka, Madhya Pradesh, Orissa and Rajasthan. It took forward the Krishi Darshan model of community TV sets, which were installed in the schools. A significant part of the programming was devoted to agriculture communication. The impact study conducted by the Planning Commission found that "gain in knowledge and attitude per respondent was the highest in the field of agriculture followed by family planning, animal husbandry and health."

SITE project continued for a year from August 1975 to July 1976. The communication process employed in the project was unique in more ways than one. Extensive time, efforts and resources were invested in it. Various mechanisms such as audience profiles need assessment, pre-testing of programmes, feedback and summative/impact evaluation were roped in.

KHEDA

The Kheda project was the precursor of SITE. It was implemented in Kheda district of Gujarat, in which low-powered transmitters were provided by the United Nations Development Programme (UNDP). A satellite earth station called Space Applications Centre (SAC) was set up in Ahmedabad. A studio was established at Pij village, and 651 televisions sets were installed at 400 villages within a range of 35 kilometres. Kheda and SITE's operations commenced in July and August 1975, respectively. The Kheda project continued long after SITE had been wound up. It went on to win the UNESCO prize in the best rural communication category in the year 1984. The true significance of Kheda lies in the fact that it was the first experiment of user-generated content achieved through analogue mode. However, in 1985, when a more robust transmitter was set up in Ahmedabad, the Kheda transmitter was dismantled and sent to Chennai for starting an entertainment channel. There were widespread protests, but these went largely unnoticed. P.C. Joshi notes, "[The Kheda Project] was wound up under tremendous pressure of the new rural middle class which was carried away by the glitter and glamour of the new television software..."

It is noteworthy that the White Revolution, which gave birth to the unique cooperative movement popularly known as Amul, took place in Kheda district. In more than one way, the Kheda project helped educate the village people to nurture a project like Amul.

Unlike Green Revolution and SITE, the People's Archive of Rural India (PARI) is a voluntary initiative. It was established by the Magsaysay award (2007) winning journalist P. Sainath. His seminal work, *Everybody Loves a Good Drought*, is a concrete example of rural reporting. Sainath is widely credited with bringing rural issues back to the public discourse. According to its website, "PARI is both a living journal and an archive. It will generate and host reporting on the current and contemporary countryside while creating a database of already published stories, reports, videos and audios from as many sources as it can. All of PARI's content comes under the Creative Commons, and the site is free to access. Anyone can contribute to PARI".

Check Your Progress Exercise 1

Notes: 1) Use the space below for your answer.

2) Compare your answers with those given at the end of this Unit.

1) What is Agriculture reporting?

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2) How agriculture journalism has contributed in the success of Green Revolution?

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3) What is PARI?

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3.3 ISSUES IN AGRICULTURE AND RURAL REPORTING

The agriculture and rural beat demands that the journalists who cover it are well aware of the issues. When it comes to agricultural issues about government

schemes, journalists usually receive a lot of literature. The journalists mustn't take the information at face value. Instead, they should engage with it critically and ask pertinent questions. Thus journalists should give enough space to the doubts, queries and untold stories, which generally do not find space in the commercially driven media eco-system. For any government scheme to be successful, the problems faced by the rural population must reach the government. Thus journalists covering this beat should create a bridge between the government and the rural populace.

3.3.1 Agriculture Technology

Modern agriculture is about better and scientific agriculture technology. As mentioned earlier, agriculture communication gained prominence after the Industrial Revolution. The new technology of tilling, sowing, reaping, and targeted chemical pesticides and fertiliser were developed to increase agriculture productivity. Changing people's habits with regard to using traditional means of farming to adopting modern technology required concerted efforts on the part of all stakeholders as well as media. This is an area on which rural/agriculture journalists should focus upon.

3.3.2 Farm distress

In the past few decades, the issue of farmers' suicide has become a much-debated topic. It has been widely reported that people are withdrawing from farming and leaving villages in search of jobs. This condition is known

as farm distress. Farm distress takes place because of various factors as given below:

- Crop failure
- Low productivity
- Natural disasters like drought and floods
- Price volatility
- Insurance failure
- Irrigation failure
- Lack of cold storage facility
- Marketing problems
- Climate Change
- Indebtedness

You can add other factors to the list. The level of distress is often different in various states due to different cropping patterns. Knowledge about the causes of farm distress is essential for balanced reportage.

3.3.3 Poverty

Poverty is detrimental to the growth of agriculture. One of the significant factors of poverty among rural populations is the lack of land reforms, apart from other issues like lack of training and proper communication process. Moreover, the development model that India has been following is not quite sensitive towards

the requirement of the rural population. There is a growing tendency amongst educated urban people to blame the poor for their poverty. The rural or agriculture journalists need to be sensitive towards the people they report about and critically engage with poverty.

3.3.4 Marketing Agricultural Commodities

Agriculture is not merely about production. It can mean the only production when it operates under the subsistence mode of economy. In the modern sense, agriculture is about strategic production relating to marketing. Therefore, market probability and potentiality drive agriculture activities. For example, cash crops like jute, cotton, sugarcane, tobacco, oilseeds etc., are entirely at the mercy of the market. If the market does not support the farmer, then farm distress becomes inevitable. In the case of food crops or even horticulture crops, the problem is different. If the market fails, the farmers can use the grains for their consumption, but they also require cash to pay back their loans. The distress eventually sets in. The storage of food crops and horticulture crops is also a challenging area. Hence, understanding the relationship between the market and farm is essential for the rural/ agriculture journalist.

3.3.5 Government Policies

We have earlier explained the symbiotic relationship between agriculture and government policies. Ideally, there should be agriculture policies that are beneficial to the farmers as well as the governments. Sometimes, the policy frameworks do not yield the expected results. Often, the benefits of policies do not reach the poor and marginalised farmers, which become the cause of farm distress. For example, maximum farmer suicides are being reported from the areas where cash crops like cotton or sugarcane are being cultivated. Therefore, to report on suicide, a rural journalist should have in-depth knowledge of government policies applicable to the area. A reporter will have to also report on the efficacy of these policies – whether they benefit the people or not. The contentious issues which plague farmers are potential areas of reporting.

Check Your Progress Exercise 2

Notes: 1) Use the space below for your answer.

2) Compare your answers with those given at the end of this Unit.

1) Define farm distress.

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2) Explain the importance of agriculture marketing.

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- 3) Why the knowledge of government policies is important for agriculture and rural reporting?

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3.4 MEDIA AND AGRICULTURE AND RURAL REPORTING

India has witnessed a massive proliferation of mass media after the implementation of neo-liberal economic order. Prior to this, the range of media in India was rather limited. However, it is a proven fact that the effects of mass communication are immense in the life of the rural population as revealed by SITE and Kheda experiments, discussed above.

In this section, we shall discuss the relationship between media and agriculture. We shall try to understand the importance of agriculture and rural reporting in a democratic set-up and explore how much space does agriculture beat get in the present day media. Media is considered the Fourth Estate. However, in the post-liberalisation era there has been a concerted effort to erase rural India from the canvas of media. It is often argued that the rural India has almost disappeared from the Indian media.

In fact, agriculture or rural issues find space in commercial or private media only in case of extreme disasters which can be sensationalised. In India, there is no media policy or framework which makes it mandatory for media to necessarily report on agriculture and rural issues. The capital city of Delhi witnessed many farmers' protests in the years 2018-19, which were largely ignored by the mainstream media. At present, we do not have a specialised beat for agriculture journalism in our media. P. Sainath notes, "Not a single newspaper in India today has a full time-labour correspondent. Labour is handled by the industrial relations portfolio of the business reporter." In such a scenario the example of PARI looks quite promising.

3.4.1 Dissemination of Farm Technology

We have already mentioned how dissemination of information regarding farm technology is a must for adaptation of that technology. Everett M Rogers, development communication theorist, termed it Diffusion of Innovation which states that unless the information of an innovation is disseminated, people will not know about it and the innovation will be of no use. Therefore, for an innovation to gain acceptance and eventually used, media has a crucial role to play. It becomes essential that an agriculture/ rural journalist is well aware of the new technological developments and also highlights the contribution of those behind the innovation. In this way, a journalist becomes a bridge between the innovator and the adopter

3.4.2 Selection of Media

Selecting the right medium and the right format for effective communication are some of the challenges for an agriculture journalist. There has been a tendency amongst media practitioners to use the media-mix approach for communication which has proved costly and less effective. Therefore, it is essential to know the media habits and utilisation patterns of the target audiences. In the good old days radio was the most used medium of mass communication in India. Since agriculture journalism began with radio communication in India, it became a natural choice of medium amongst journalists. Moreover, AIR provided ample space for agriculture and rural communication. Today online platforms have a huge potential to engage with the farmers in a far more interactive and creative manner. However keeping in view the poor literacy level of rural population, it is not possible to completely do away with the traditional media especially radio

Activity

Follow the news of farm distress in a regional newspaper of your choice. Analyse if the selection of media was appropriate in the light of topics and subjects covered.

3.5 WRITING FOR AGRICULTURE AND RURAL COMMUNICATION

In previous units we have explained in detail writing and reporting techniques for various media. All these techniques are required for agriculture and rural communication as well. Writing stories and news on agriculture and rural issues is not much different except that the technical terminologies should be written in a more comprehensible way. A rural journalist is required to write a story in a simple and lucid style. The story must be balanced, objective and nuanced. Rural journalists should also ensure that facts are never compromised and remain sacrosanct.

There are no specific formats as such for writing on agriculture or rural issues. However, interviews, features, success stories and hard stories backed up with data have been extensively used. Popular articles by experts in agriculture and extension education are also published. The selection of format needs to be made with care in the light of the subject and themes that have to be communicated. For instance, to communicate the significance of using a particular variety of seed, interview technique will be more suited. The success story of a farmer who has already reaped the benefits should be interviewed to inspire other farmers for using this variety of seed.

While writing stories on rural agricultural issues, you are likely to come across a great deal of data which needs to be communicated to the target readers. The National Family Health Survey, census report, reports on poverty and farmers' suicide, reports on agriculture production etc. are replete with data. These reports and data are generally available online. It is a proven fact that agriculture labourers who do not own land titles are generally missing from many official/government data. Similarly, women farmers do not figure adequately in the government reports and data. For instance, much of the farmers' suicide cases are also not properly reported. Therefore, the real challenge is to gather authentic data on such issues and report them.

Check Your Progress Exercise 3

Notes: 1) Use the space below for your answer.

2) Compare your answers with those given at the end of this Unit.

1) What is the role of media in agriculture communication?

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2) What is the importance of selecting proper format for agriculture reporting?

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3.6 LET US SUM UP

In this unit the role of media in dissemination of information of agriculture and rural issues was described. We started with the origin of agriculture journalism and how it was related to the Green Revolution in India. It was explained that as part of development communication, agriculture reporting is positive, practical and pragmatic aimed to contribute in the well being and development of people. We also discussed various initiatives like SITE, Kheda and PARI in agriculture and rural communication.

Some pertinent issues such as farm distress, poverty were explained and the need for focusing them as well as on agriculture technology, marketing and government policies in reporting were highlighted. It was observed that radio and television are effective in reaching out to the farmers. Lately, online platforms have also been found useful in engaging with farmers in an interactive manner. Some specific requirements for writing on agriculture and rural areas were also touched upon.

We hope that after going through this unit, you would understand the responsibility of an agriculture and rural journalist and use appropriate language, words and visuals in a news story. The foremost requirement in a rural journalist is the sense of responsibility which should get reflected in your reports for the benefits of rural and poor farmers.

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3.8 ANSWER TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress Exercise 1

- 1) Agriculture reporting refers to the reporting of the processes of food production and distribution. It essentially deals with the political economy of agriculture sector. Agriculture reporting is meant to empower farmers by informing them about the best agriculture methods and latest technological innovations.
- 2) Green Revolution was initiated to increase the production of food- grains through the use of new scientific technology. Indian farmers who were using old, traditional farming methods needed to be apprised of the benefits of new technologies. Agriculture journalism contributed in disseminating the information regarding new developments in the domain of agriculture far and wide, thus helping the farming community.
- 3) PARI is a voluntary organisation founded by the Ramon Magsaysay award winning journalist, P. Sainath. PARI is widely described as a journal and an archive of rural affairs. It generates tropical and relevant journalistic stories from the rural sphere of India which is fast disappearing from the mainstream urban media.

Check Your Progress Exercise 2

- 1) The distress faced by the farmers' community is known as farm distress. It takes place due to natural as well as artificial factors. Natural factors include conditions like drought and floods. On the contrary artificial factors include misplaced and faulty government policies and their implementation, manufactured price volatility, poor infrastructure, uncertain marketing etc.
- 2) Agriculture marketing refers to the services pertaining to varied agriculture products. It essentially deals with the key commercial aspects of agriculture: Price, Product, Promotion and Placement.
- 3) Government policy framework directly effects the rural livelihood. It shapes the future of the rural population. Therefore, agriculture/ rural journalists need be well educated with regard to the government policies and schemes pertaining to agriculture sector. These should be adequately reflected in their journalistic work.

Check Your Progress Exercise 3

- 1) Mass communication is one of the most effective means of reaching out to the farmers and rural population. Radio and television have wide outreach and access and are considered to be effective media for the rural population. Online platforms have also been found effective in engaging with farmers in an interactive manner.
- 2) The selection of format needs to be made with care in the light of the subject and themes that have to be communicated. For instance, to communicate the significance of using a particular variety of seed, interview technique will be more suited in which a farmer who has already reaped the benefits from it may be interviewed to inspire other farmers for using this variety of seed.

UNIT 4 SCIENCE & TECHNOLOGY ISSUES

Structure

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4.0 INTRODUCTION

Science and Technology (S&T) are increasingly determining the development and lifestyle of human beings on this earth. The science includes aspects such as technology, nature and medicine. Science has increasingly played an important role in moulding human society by its innovations, theories and discoveries, and is expected to do so in more ways in future as well. Therefore reporting about such essential aspects of our lives becomes essential and a fascinating endeavour.

Science, Technology and mass media have a long-standing relationship with each other. Mass media makes scientific knowledge citizen-friendly by keeping the public informed of the latest developments in Science and Technology. Great deals of scientific information common people have comes from mass media sources. Science writers and journalists shape mass consciousness about science-related events (Nelkin, 1995). Mass media is increasingly being viewed as

necessary for the science and scientific community, which recognizes the power of media for its ability to facilitate interaction between the scientific establishment, public, interest groups and policymakers. Mass media, in this sense, has become the primary channel through which scientific notions and latest technological progress can be conveyed to the people.

4.1 LEARNING OUTCOMES

After studying this Unit, you should be able to:

- explore the relationship between science, technology and mass media;
- discuss major events in the history of science journalism;
- outline major forms of science and technology reporting;
- explain some contemporary scientific and technological developments;
- analyse reporting on controversies regarding science and technology; and
- explain issues related to science and technology reporting.

4.2 CONTEMPORARY SCIENCE & TECHNOLOGY DEVELOPMENTS

Scientific research and technology is playing a major role in empowering and uplifting masses all across the developing world. From food security for millions of people to computational tools and space technology, science and technology is becoming a primary factor in determining economic growth. The advanced countries also exercise power through science diplomacy. It is argued that scientific knowledge and technological innovations will play a key role in attaining the goal of sustainable development in emerging economies like India.

By means of its acquired scientific standing and technological progress in Post-Independence era, India has been successfully dealing with the age-old problems of food insecurity, poverty and high mortality rates. The country is recognized today as one of the emerging global scientific power.

Contemporary developments in science and technology are radical, disruptive and possess far-reaching effects on humans and other living systems on this earth. Some of these technologies and developments will also have socio-political and socio-economic impacts on cultures across the globe. Hence as science journalists, it is important for you to keep a tab on such developments. Following is a brief description of few such technological developments.

4.2.1 Stem Cell Research

Stem cells are living cells that can change into other types of cells. Each multi-cellular organism such as animal and plant has stem cells. In mammals such as humans two types of stem cells are embryonic stem cells and adult stem cells. Embryonic stem cells are found in early stage of human embryos and adult stem cells are found in various tissues of adults. These stem cells operate as repair system for the body. While the embryo is still developing into a fetus, stem cells are able to differentiate into all the specialised cells such as outer skin and various internal organs. There are various medical applications of stem cells. For example,

adult stem cells are used in therapies such as bone marrow transplantation. Stem cells are now artificially cultured and transformed into any types of cell such as muscles or nerve cells. Embryonic stem cells can be taken from umbilical cord blood immediately after a child's birth. It is expected that stem cell therapy (also called as cell replacement therapy or regenerative medicine), would in future help in treatment of diseases such as stroke and brain injury, baldness, heart diseases, deafness, diabetes, cancer, wound healing and male infertility.

Stem cell research initially caught a lot of media attention because of ethical and political controversies tied to it. The main controversy was related to development, use and destruction of human embryos for the sake of using embryonic stem cells.

4.2.2 Genetically Modified (GM) Food Technology

Genetically Modified (GM) technology allows scientists to transfer genes with specific traits between species using laboratory techniques. Some examples of GM application in plants and animals are bananas that have vaccine effect in them, fish that matures quickly, fruits that yield years earlier and plants that can produce new plastics. Application of GM technology in plants includes nutritional enhancement, disease resistance and bio-fuel efficiency. As they are referred to sometimes, GM crops or transgenics contain genes from another species, which gives them some specific properties like resistance to a particular disease. The principal transgenic crops grown commercially in some countries include soybean, corn and cotton. India has allowed only GM cotton for commercial cultivation in the country.

There is an ongoing debate regarding the potential health risks and environmental side effects, and the potential benefits of GM crops. Controversy and public concern around GM food and crops are related to human and environmental safety. What impact will they have on the environment and biodiversity and human health?

In India, reporters widely covered GM debate when Bt Brinjal (a GM variety of brinjal) was approved for commercial cultivation in the country. Due to resistance from civil society and few other organisations and its subsequent coverage in mass media, a moratorium was finally put on it by the Government of India.

4.2.3 Artificial Intelligence

Artificial Intelligence (AI) is a way of making computer, robot or a software think intelligently just like humans do. AI is also the study and design of intelligent machines, which have ability to learn from experience, adjust and respond to new inputs and perform human-like tasks. The science of AI includes areas such as computer science, biology, neuroscience, psychology, linguistics, mathematics and engineering. Popular examples of AI include chess playing computers and self-driving cars.

AI is among the most debated technological issue in media today. The questions which are raised in the context of emerging intelligent machines include those related to control. How do we control such a complex intelligent system is what worries thinkers and intellectuals. It is argued that at some point of time in future, machines and robots will become so sophisticated and clever that human being

will no longer be the most intelligent beings on the earth. At that point of time, who will control whom will be a big challenge. Another problem with AI is that it is risk prone and how would this risk be managed or controlled is also a big question.

4.2.4 5G Wireless Technology

5G or ‘fifth generation’ represents next generation mobile networks. 5G is expected to revolutionise the way we communicate with each other and the way we carry out day-to-day business. 5G is expected to go beyond the limitations of 4G, 3G and 2G, offering exciting possibilities and a number of other applications. Some of the features of this new mobile technology are as follows:

- It will offer very high mobile and wireless internet speeds while at the same time expanding connectivity to a large geographical area. It is expected that 5G would be 200 times faster than its predecessor 4G.
- It will have better quality, voice and video calls.
- 5G would help in creation and expansion of advanced technologies in sectors such as transport (e.g. driverless cars), health (e.g. telemedicine), smart energy networks, smart agriculture and smart hyper-connected cities.
- 5G networks would be less energy consuming thus offering more battery life to handsets, tablets and other connected devices.

4.2.5 Internet of Things

The term ‘Internet of Things’ (IoT) is a general term which includes everything connected to the internet ranging from smart watch to washing machine to the kitchen oven. It refers to the eco-system of smart things both indoor as well as outdoor, which are connected to each other as well as to internet, thus exchanging data and information. The devices are fitted with sensors and software and thus can easily identify and connect wirelessly to each other. Journalists use this term quite often to point to the next connectivity revolution.

4.2.6 CRISPR Gene Editing Technology

Scientists say that once we understand the DNA sequence in cells and acquire tools and techniques that would allow easy manipulation of that sequence, we can create a revolution in biological world. CRISPR is a powerful technology that would allow scientists change the sequence of DNA in cells in a precise way to correct mutations that otherwise cause disease. CRISPR would enable us to do lot that was impossible to do in the past. It is an exciting technology that is going to change a lot of things in human health sector as well as in agricultural applications. CRISPR technology is being applied in food and agricultural industries to engineer probiotic cultures and to improve crop yield.

There is a word of caution in respect to the use of CRISPR technology and its applications such as with that of human embryos as it might lead to designer babies with desirable physical traits and talents. Some are also concerned about the experiments with DNA of disease spreading insects or to genetic enhancement crops or livestock because we are not aware about their unintended impacts on environment or human health.

Check Your Progress Exercise 1

Notes: 1) Use the space below for your answer.

2) Compare your answers with those given at the end of this Unit.

1) How mass media helps in legitimating scientific knowledge?

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2) What is Artificial Intelligence (AI)?

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4.3 SCIENCE AND TECHNOLOGY REPORTING

Science and Technology reporting is among major areas of reporting in the contemporary media landscape. Newspapers dedicate special pages and supplements for covering developments related to Science and Technology. Science is now gaining more and more prominence in mass media with more scientific events and technologies being covered by reporters on a daily basis.

4.3.1 Evolution of Science and Technology Reporting

We find that Science and Technology stories have always appeared in mass media. In earlier times, articles and descriptions were written by scholars explaining natural phenomena and basic scientific concepts. The aim was to make people understand the basic notions of science and scientific phenomenon. By the nineteenth century, several popular science magazines such as Scientific America and Popular Science were started. News editors began to reprint science lectures and published scientists' descriptions of natural phenomena such as meteor showers. As science and technology intruded into our lives and science became more global, it gained more prominence, and media began to cover and report scientific events.

Among the first articles written by a science correspondent was 'A Bale in the Bay of Biscay'. It was published in The Times in 1871. Science journalism took shape in the world with the appointment of James Crowther as the science correspondent of The Manchester Guardian in 1928.

The first science news service was founded in the US in 1903. Around the time of the Second World War and other scientific and technological developments, especially in war technology, science journalists began to be seen as

communicators, observers and critics of scientific developments, with their political implications. The technology of warfare, discoveries of planets and breakthroughs in medical science were easy for journalists to cover and write about because media editors readily accepted such news stories.

By the 1930s, some journalists were starting their careers in science reporting. They were becoming critical of the practice of science and journalism and social issues arising out of science and technology. In 1934 the National Association of Science Writers was formed in the US.

The European Union of Associations of Science Journalists (EUASJ) was founded in 1971. In 1974 science journalists from nine European countries met in Salzburg-Austria to discuss their role in the modern world. This conference was convened by the newly formed European Union of Associations of Science Journalists (EUASJ) and resulted in the Salzburg Declaration on Science Journalism. The conference, as well as the declaration, was the first of its kind in the world. Another milestone was achieved when The New York Times created a separate science section in 1978. By 1987, the US alone had some 147 newspapers with at least a weekly science page.

In India, modern science journalism evolved during the 1960s and 70s with the emergence of popular science magazines. The majority of mainstream Indian newspapers started dedicating columns or pages to science and technology news only in the late nineties. Science journalism saw its peak in the 60s and 70s, and then it decelerated during the 80s and 90s. It started to pick up with major national and regional dailies providing regular and dedicated space to science and technology. Lately, online news platforms have started employing science editors who have science degrees - unlike political reporters, who could be from any stream. Online platforms now have dedicated portals for science news.

4.3.2 What is Science and Technology News?

Science and technology news includes the account of scientific and technological developments which happen in the world around us. It is about reporting the conferences, advancements and discoveries from the field of science and technology or even infectious disease outbreaks, along with informing and explaining about their overall implications, be it social, political, economic or health. Nwabueze defines science and technology reporting as ‘the application of journalistic principles in conveying information about science and technology issues, topics and developments to the public through the mass media’ (Nwabueze, 2009; 324). News stories generally have short paragraphs and sentences. They are written and organized in a manner that ‘who’, ‘what’, ‘when’ and ‘where’ are mentioned in first paragraph of the news story.

4.3.3 Features of Science Reporting

Many breakthroughs and increasing relevance of science in today’s modern world have made science and technology reporting a specialized area in media. Science reporting is unique and differs from other journalistic beats. Science stories are usually shorter than other types of stories. Following are some special features:

- Science journalism is about reporting complex scientific issues to general audience.

- Science and Technology reporting sometimes involves reporting about incidents, events and concepts which are abstract, complex as well as difficult to understand for a layman audience. Hence a reporter should have a clear understanding of what s/he is going to write about and how to put that in layman's language.
- Science and Technology reporting is about human progress and how the future would look like. So a reporter should be far-sighted enough as to visualize the future changes and impact.
- Science reporting also involves questioning the statistics; investigate scientific misconduct, conflict of interest and ethical breaches.
- Science is never static and new knowledge comes each day. A good science journalist should constantly keep him/herself updated.

The art of good science reporting demands two minimum requirements: basic knowledge of science and training in journalistic skills about how to cover science and technology issues. A good degree in science is no doubt helpful. However, some popular science journalists and science writers such as Carl Zimmer and David Quammen had no formal training in science. Great science writers such as Walter Sullivan of the New York Times and many other journalists and writers of science belonging to leading newspapers of the world were self-taught. However, one thing was common to all of them they had a passion for science and technology.

Check Your Progress Exercise 2

Notes: 1) Use the space below for your answer.

2) Compare your answers with those given at the end of this Unit.

1) What is science and technology news?

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2) What are the unique features of science and technology reporting?

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4.4 BECOMING A SCIENCE REPORTER

The following steps will be useful for science writing or reporting.

4.4.1 Steps in Science Writing

A good science story can be about an issue, a new research study, new technology or some new theory. It can even be something closely related to science such as some policy or a specific law related to any aspect of Science and Technology. It

may also warn us about the potential threat posed by a new technology or give a detailed account of a scientific controversy.

4.4.2 Sources of a Science Story

There are various sources of Science and Technology news. The following are some of the common sources:

Personal Observation

A reporter can identify some new development world of science. S/he may observe some new trends in his/her own surroundings which are worth reporting.

Scientific Journals and Publications

Such publications are a ready source of information about current research happening in field of Science and Technology. They contain results of various scientific studies done by scientists in laboratories around the world. The reporter should first understand them and then write a story based on the information in there publications.

Visit to science research institutes

Visit to such places can always yield newsworthy stories.

The Internet

Internet is the most accessible source of science news. A wide range of databases and websites on the internet are vital sources of scientific information.

Science Conferences and Conference Proceedings

Scientists of leading research institutes organize press conferences to make announcements about the new innovations and work done by them. Every year conferences are held where scientific community belonging to a specific discipline, meets and discusses new findings and current status of research in the area. Such conferences are important sources of news in specific areas such as nanotechnology, plant bio-technology and space science.

Interviewing a scientist

Interview is the exchange of information between a reporter and a source/scientist. Science and Technology reporting interview is a very important tool for digging out facts and figures which otherwise might be hidden from the public eye. You will read more about conducting interviews in sub-section 8.5.2 of this unit.

4.4.3 Tips for Writing Good Science Stories

Understanding the details

One must have a proper understanding of the concept as well as the context for writing a news story accurately. A reporter doesn't have to know as much as a scientist, to report any scientific issue or development. Only a basic understanding is required to contextualize the story and to supplement the facts with their own knowledge so that general readers can understand.

Any science and technology issue will have human application as well as impact. For every story, a journalist should ask, how this research or technology would affect my readers. Look for a human interest element in the story.

If a research or news has no impact on everyday life of people (such as astronomy), try to make it relevant for the readers. Sometimes scientific research is too complicated to understand. If you have doubts, on any point, visit a knowledgeable scientist and ask him/her to explain the details until you have understood the concept.

Use metaphors and analogies to explain difficult concepts

Metaphors and analogies are an important part of science writing. They help in creating images from everyday life and culture, which may be used to explain concepts and thus make science easier to understand. For example, atmosphere acts as a greenhouse.

Don't exaggerate and sensationalize the claims

Be careful with scientific claims. People generally trust such claims when they are put by a science reporter. Never try to exaggerate and make sensational claims which may not be true. Sensationalism is when a reporter puts a story or exaggerates any claim, in such a strong manner that it affects people. Similarly reporter should not say beyond what is true, to draw the attention. Sensationalising science and exaggerating the facts may lead to false hope or fear among the public.

Check and double-check your facts

Presenting accurate and precise facts is a must in science news reporting. Always double-check your facts. If doubts remain, check with an independent expert. Ask the question- what does the data in the study actually say or highlight? If you don't understand the data fully then you cannot competently report on science.

Be objective and balanced

Science and Technology reporter should present all sides of a story to readers, to enable them to make informed decisions. The story should be balanced without putting one's own prejudices and biases. However, balancing doesn't mean giving all viewpoints equal weight.

Write well, avoid jargons

Include those details in your story which readers want to know. Write crisp, clear and in logical order so that story has a flow. Stick to one idea in each paragraph. Avoid jargons and technical terms and instead use layman language.

Provide context and background details

The task of a science reporter is to provide relevant details and history regarding the study and also to place the developments in a proper context.

If you find the answer to your question from the author unsatisfactory, check with another expert and get feedback

4.5 FORMATS OF SCIENCE REPORTING

Science reporting and writing can take many forms and shapes depending upon the context, target audience, source of story and the purpose of writing. The following is a brief description of major forms of Science and Technology reporting.

4.5.1 Based on Research Paper

Most of the science stories you come across in mass media are based on the latest research published by accomplished academic and research journals. It is possible to write a good story based on the contents of a research paper. Following are the guidelines for covering a research article or paper:

- ***Reading the paper***

Do an exhaustive reading of the paper. The abstract gives the major points of the work. Read introduction, the discussion part and the conclusion. These provide context as well as observations and the future implications of the study. Read the data and statistics carefully before you draw any conclusions. Consult internet or read supporting material for understanding the complex concepts.

- ***Check the conflict of interest***

Check who is funding or sponsoring the project. This is generally mentioned at the end of a paper.

- ***Look at previous studies***

Look at the previous studies done on the same subject or theme in order to get a full picture. Generally information about previous studies is given in 'literature review' in a research paper.

- Put most important or major finding of the study first and then go on elaborating and giving context, implications etc.

4.5.2 Interviews

Interview with a scientist makes the story more interesting and adds to its impact. Get the authors to explain the results and their implications. Try to understand what the results mean in plain English. Catch a dominant line from the interview which you can put in the story as a direct quote in the story. There are various types of interviews that a science and technology reporter may use to acquire the required information:

Exclusive interview

This kind of interview is given to a reporter by a scientist or an expert, without any invitation extended to any other reporter. It takes place with prior consent of the interviewee and the information disclosed during the interview is known only to the reporter who conducts the interview.

Telephonic interview

This interview is conducted on the phone and is not face-to-face. The main advantage of this kind of interview is that it saves a lot of time and is also cost effective but may not yield detailed in-depth information.

Personality interview

This interview is conducted by reporter with a view to acquiring information directly from the person whose research work and discoveries the reporter wants to cover or write about. The person could be a celebrity or a well-known scientist.

Personality interview like exclusive interview is also fixed in advance with prior appointment. The reporter should do proper homework regarding the life and work of the scientist.

On-the-spot interview

This interview is conducted on the scene of an event - it could be done on the sidelines of an ongoing conference or even in a research laboratory where an important discovery is being made or some novel experiment is under-way.

Vox-Pop or voice of the people

It is conducted when a reporter wants to generate public opinion on an issue of public interest. e. g. it could be about some new technology, say mobile technology; pandemic like Corona virus; or a new medicine.

News interview

The aim of this interview is to gather background information that will help explain a news story.

While interviewing a scientist, a reporter should follow a proper interview technique:

- Introduce yourself adequately
- Be friendly, courteous and sympathetic. Generally scientists are media shy and don't speak with journalists. Make the scientist comfortable so that s/he opens up.
- Begin the interview with light and interesting questions so that s/he becomes comfortable with you. Drafting questions beforehand may help you to be more structured and systematic in the interview.
- Listen carefully to whatever is said and when needed pose questions arising out at the discussion. Take notes as you interview. You can also use a recorder or phone but first ask for the permission.
- Never agree to ask pre-agreed questions.
- Keep on asking questions until the issue or the concept is fully clear to you.

4.5.3 Press Conference

Scientific conferences are great places for meeting many scientists and finding new ideas. They are major scientific events as they bring together many renowned experts of a particular field at the same place, who may otherwise be difficult to contact. The experts discuss new ideas, present novel results and conclusions, and sometimes also make new recommendations with implications for society. Conferences enable journalists to know the latest happenings in the field. Doing some basic homework is essential before covering a conference. With a little background reading on the conference topic/theme, a reporter will be in a better position to tell which scientist is presenting what and is there something new in what he presents. As a reporter you should:

- Watch the presentation carefully, listen to what the presenter says and at the end ask questions.
- Observe and watch what important trend is emerging from this conference discussion.

4.5.4 Long-form Science Writing

A science feature is a long form of science writing with more depth than a news story. It is a form of non-fiction writing which gives background information and underlying causes on certain major scientific events, or recently done research. They usually have an interesting introduction that grabs the reader's attention. Science feature article normally presents newsworthy events and information related to any aspect of Science and Technology through a narrative story. For example a feature on the Mars mission by Indian Space Research Organisation (ISRO) which has either happened very recently or is being planned in near future, or a feature on some research breakthrough by scientists at Indian Institute of Science (IISc), Bangalore. Features are generally written in active voice.

A science feature needs to have voices of those involved in the issue. However, every aspect of an issue cannot be covered in one single feature. A science feature includes the element of factual reporting and some added story elements of imaginative writing and interpretation. A science feature gives readers factual information or interpretation of some technology in an interesting form. But it goes beyond mere interpretation of facts by adding relevant studies and other information. Features generally require the writer to apply his/her imagination and interpret facts in a way which interests the readers. While writing a feature, one should put the major and interesting information in first half of the story and then provide context, background and implications.

Feature stories vary in length but generally magazine features are longer than newspaper features because of availability of more space in magazines. A science feature like other features is seldom written in traditional inverted pyramid form. It should flow smoothly in a well-planned order.

4.5.5 Scientific and Technological Controversies

Scientific controversies grab attention of media. Often editors allot prominence to specific scientific controversies by publishing them on the front page; while in other cases, such stories act as a tool for informing the public and creating awareness among them on sensitive issues. Informed coverage of scientific controversies not only benefits public but may also lead to a change at the highest levels of public policy. For example well-articulated media coverage of health hazards of traditional wood stoves for cooking led to the programme of 'smokeless chulas' in India.

At times scientific controversies erupt, when there is disagreement among a group of scientists regarding any aspect of scientific research or technological development. As scientific research progresses, new knowledge produced can cast doubt on existing knowledge or application, which was based on previous understanding. This becomes controversial at times. For example, genetic modification (GM) technique, hailed by majority of scientists as a tool for growing better crops became a highly controversial issue after the report in journal Nature on the death of monarch butterfly larvae, fed on GM corn.

Controversies on scientific issues come to public notice when they get a place in media. In any scientific or technological controversy there is more than one side of the story. When two sides of scientific explanation are pitched against each other, the job of media is to present a non-prejudiced and balanced picture of the

story, giving voice to the stakeholders. For example, in GM food controversy there is a community of scientists which supports it and on the other hand there is another community of scientists who opposes. So, the reporter has to do a tightrope walk, and has to balance it without leaning on any one fulcrum and staying neutral on all grounds.

Scientific controversies are multi-dimensional and have implications in politics, religion and industry. For example, although the core of climate change is atmospheric science and geography, yet actors in this controversy include politicians, industry, activists, non-governmental organisations and common people.

Almost every new technology which comes up these days has its own share of controversy such as bio-technology, heart transplants and nanotechnology. Controversies don't settle down quickly. Keep monitoring them by visiting websites in order to know how these controversies are being covered by media of other countries. One should not write about a controversy unless s/he has double-checked the facts.

4.5.6 Covering Health

For many media outlets today, the bulk of what consists of science news writing is mostly about medicine and health. Health journalism is about covering medical and health care related issues. Coverage of health related issues change in knowledge at the individual level, and initiate agendas and actions at the institutional and policy level. News media plays a leading role in defining priorities in health sector and also disseminating new research findings to the public. Mass media especially the press is seen as an integral source of public policy because of the way it frame the issues. Effective health reporting is also needed to promote awareness and understanding of health-related issues.

Health is an important journalistic beat. This beat touches on economics, politics, public welfare, medicine, resource allocation, fake drugs, marketing, sales practice, consumer law and regulation. Health reporting may also deal with unethical practices in health sector which range from unholy nexus between drug manufacturing companies and doctors to non-availability of clinical trial data in public domain. Health beat is one of the most important and complex beats with a wide range of issues to cover.

4.6 CHALLENGES IN SCIENCE AND TECHNOLOGY REPORTING

Research is the essence of science and technology reporting, and it comes with its own set of challenges. Unlike other areas of reporting, Science and Technology reporting needs lots of background work and study. Understanding the data, and converting the data into user-friendly media format is the moot point in Science and Technology reporting. Since information has to be dug out from various research organisations science reporting needs basic disciplinary knowledge as well as primary investigative skills.

Journalists often face pressures that confuse them about competing scientific claims in terms of their credibility. Determining how much coverage to give to

different sides in a scientific debate requires considerable expertise on the issue at hand. New knowledge and disciplines emerge almost on a daily basis. A science journalist may have sound knowledge about one area but may have very little knowledge about other areas. In such cases s/he needs to update oneself about these new developments to report them accurately and in a proper context.

Sometimes covering a scientific issue or controversy means questioning big corporations and multinational companies, which indulge in corrupt scientific practices or even unearthing undisclosed ties of research institutes with some multinational companies. The job of a science reporter is to empower the common man and portray the truth. Science journalist should look beyond wonders, theories and data. S/he should observe the people involved in science and see whether they have any conflict of interest. S/he should also analyze the power structures to see as to who is included and who is excluded or marginalized in the world of science.

Check Your Progress Exercise 3

Notes: 1) Use the space below for your answer.

2) Compare your answers with those given at the end of this Unit.

1) What are the basic requirements of a science journalist?

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2) What should be the role of journalist in case of a scientific controversy?

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4.7 LET US SUM UP

In this unit various facets of science journalism were explained. We discussed some contemporary developments in the world of science and technology such as 5G technology, stem cell research, artificial intelligence, internet of things and CRISPR gene editing technology.

We briefly discussed history of Science and Technology reporting and described science and technology news and the art of science news reporting. The unit gave an overview of major sources of science news and how good science stories are written. We also discussed some formats for science reporting and how Science and Technology related controversies should be reported.

4.8 REFERENCE AND FURTHER READINGS

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4.9 ANSWER TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress Exercise 1

- 1) Mass media plays an important role in legitimisation of scientific knowledge. It not only helps shape public's perception of science but also facilitates a dialogue and interaction between scientific community and other social group stake-holders.
- 2) Artificial Intelligence is a method by which computers and other such machines or a software programme is made to think intelligently just like human beings. The machines and robots with artificial intelligence are able to learn from the experiences, adjust and respond to external stimuli and thus perform many functions as human beings do.

Check Your Progress Exercise 2

- 1) Science and Technology news is about reporting new technology, inventions and discoveries in the field of science, and technology and informing about their possible implications on human beings and environment.
- 2) Science and technology stories are usually shorter than other types of stories. It deals with complex scientific issues such as theories and concepts. It involves interpretation of a lot of statistics and complex data and it also includes investigation of scientific misconduct.

Check Your Progress Exercise 3

- 1) A science journalist should have basic knowledge of sciences and journalistic skills. In fact some of the famous writers and science journalists have no formal qualifications and training in science and technology. Good science reporting is more about passion and motivation.
- 2) In the case of a scientific controversy, the job of science reporter is to empower the common man and portray the truth. Science journalist should observe the people involved in science and see whether they have any conflict of interest. S/he should also analyse the power structures to see as to who is included and who is excluded or marginalised in the world of science.